Compute Cell Scheduling Records

This document records how tasks were scheduled on small compute clusters. The workload and the machine failures are real records from a production compute cell of about 12,500 machines (the Google cluster trace of May 2011, clusterdata-2011-2, CC-BY 4.0): every task's submission time, job, priority, CPU and memory request and run time, and every machine's capacity, failure time and return time, are taken from that trace (ids are renamed). Where and when each task ran in the small cluster is not recorded in the trace: it is computed from those records by the scheduling policy stated below, applied exactly.

The document is divided into chapters titled 'Cell window <NN>'. Each chapter is an independent run of one small cluster over a window of 1200 seconds. Questions refer to a chapter by that title. Times are whole seconds from the start of the window (time 0); the window covers times 0 to 1200.

Each chapter has four sections. 'Machines' lists the cluster's machines with their CPU and memory capacity, the time a machine failed (if it did) and the time it returned (if it did). 'Task log' lists, in time order, every event of every task from time 0 to the window end: submit, start, finish, evict and fail, with the task's job, priority, CPU and memory request, run time, and the machine involved. 'Events' lists machine failures and returns, evictions and restarts, and the longest waits. 'Queue at window end' gives the state after the events at time 1200: the tasks still running (with their remaining run time) and the tasks still waiting, in queue order.

Tasks. A task belongs to a job and has a priority from 0 to 11 (higher is more important). Priorities 9 to 11 are the *production* tier and priorities 0 and 1 the *free* tier. Each task requests an amount of CPU and of memory, which it holds on one machine while it runs, and has a run time: once started it runs for exactly that many seconds and then finishes, unless it is stopped first. The run time is given on every row of the task log, starting with the task's submit row.

Scheduling policy. Every window starts with all machines empty and no task waiting. At every time at which something happens, in this order:

1. **Finishes.** Tasks whose run time is over finish and free their CPU and memory.

2. **Failures.** Machines that fail at this time go down. Each task running on such a machine stops (a *fail* event) and goes back to the queue.

3. **Returns.** Machines that return at this time come back, empty, with their capacity.

4. **Submissions.** Tasks submitted at this time join the queue.

5. **Scheduling round.** The waiting tasks are taken one at a time in *queue order*: higher priority first, then earlier submission time, then lower task id. Each task, in turn:

a. **First fit.** The task starts on the first machine, in machine id order, that is up and whose free CPU and free memory are both at least the task's requests (subject to the reservation rule in c).

b. **Preemption.** If it fits nowhere, no task holds a reservation yet in this round, and it is a production task, it may evict free-tier tasks from one machine. It looks at the up machines in id order. On each it considers the free-tier tasks running there, lowest priority first, then the most recently started first (then the higher task id first), and evicts them one by one in that order until it fits; it takes the first machine where this works, evicting only as many as needed there, and starts on it. If no machine works, nothing is evicted. An evicted task goes back to the queue at once (it may start again later in the same round); it loses its progress and will run its full run time again.

c. **Reservation and backfilling.** The first task in the round that can start neither by a. nor by b. gets a *reservation*: the earliest time at which it could start, computed from the run times of the tasks now running (assuming no new submissions, failures or returns), on the first machine, in id order, where that earliest time is reached. Every later task in the round may still start by first fit, but on the reserved machine only if it will finish no later than the reserved time, or if its requests fit in what the reserved machine will still have free at the reserved time after the reserved task's requests (and after other such starts on it in the same round). On the other machines it starts as usual. Later tasks do not preempt. A task that would not fit on any up machine even if that machine were empty gets no reservation and simply keeps waiting; the next task that cannot start may then get the reservation.

Tasks stopped by an eviction or a failure keep their original submission time (and so their place in queue order) and start again from the beginning. A task never moves while it runs.

Order of rows. Task log rows with the same time follow the steps above: finishes (by task id), failures (by machine, then task id), submissions (by task id), then the scheduling round's starts and evictions in the order the round makes them (each eviction just before the start that caused it).

Terms used in the Events section and in questions. A task's *wait* is the time from its submission to its first start. A task is *restarted* when it starts again after an eviction or a failure. The number of *evictions* counts evict events (one per evicted task). 'Running on a machine at time t' means running after all the events at time t.

The cluster and the workload. Each window has 8 machines, ids M01 to M08. Each is modelled on a real machine of the cell (one of them is a machine that failed during the window in the trace) and has 1/8 of that machine's recorded CPU and memory capacity. A failed machine returns at the time the trace records it coming back, if that is within 6 hours; otherwise it does not return. The tasks are a sample of the real tasks submitted in the cell during the window, chosen by this fixed rule: jobs submitted in the window are taken in a random order, first up to 6 production jobs, then free-tier jobs until the offered load reaches 54%, then other jobs until it reaches 90% (a job that would push it above 95% is skipped). A job contributes its first 4 tasks (16 for a free-tier job) that finished normally in the trace and fit on at least one machine of the window. Offered load = the larger, over CPU and memory, of the sum of request x run time inside the window over all tasks, divided by the cluster's total capacity x the window length.

After the window. For questions about what happens after time 1200: no new task is submitted and no machine fails after the window; a machine that is down returns at its listed return time (if any); the policy continues unchanged from the state given in 'Queue at window end'.

Units. Times and run times in seconds. CPU and memory in units of 1/100,000 of the largest machine in the original cell (the trace's normalised values, rounded to whole units).

Chapters in this document (itops_d01): Cell window 04, Cell window 07, Cell window 11, Cell window 18, Cell window 22, Cell window 25.

Cell window 04

Times 0 to 1200. The cluster has 8 machines. 102 tasks of 37 jobs are submitted during the window.

Cell window 04: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	3125	3123	-	-
M04	6250	6244	-	-
M05	6250	3116	-	-
M06	6250	3116	-	-
M07	6250	6244	-	-
M08	12500	12500	428	never

Cell window 04: Task log

time	task	job	priority	CPU	memory	run time	event	machine
293	T001	J01	0	6250	466	32	submit	-
293	T001	J01	0	6250	466	32	start	M01
296	T002	J02	1	1250	1590	309	submit	-
296	T002	J02	1	1250	1590	309	start	M02
297	T003	J03	9	6250	10340	587	submit	-
297	T003	J03	9	6250	10340	587	start	M08
300	T004	J04	8	6250	4773	569	submit	-
300	T004	J04	8	6250	4773	569	start	M04
322	T005	J05	4	1250	466	202	submit	-
322	T006	J05	4	1250	466	202	submit	-
322	T007	J05	4	1250	466	205	submit	-
322	T008	J05	4	1250	466	205	submit	-
322	T005	J05	4	1250	466	202	start	M02
322	T006	J05	4	1250	466	202	start	M02
322	T007	J05	4	1250	466	205	start	M02
322	T008	J05	4	1250	466	205	start	M02
325	T001	J01	0	6250	466	32	finish	M01
341	T009	J06	8	6250	4773	1085	submit	-
341	T009	J06	8	6250	4773	1085	start	M07
356	T010	J07	0	6250	1244	468	submit	-
356	T010	J07	0	6250	1244	468	start	M01
357	T011	J08	0	6250	1244	120	submit	-
357	T011	J08	0	6250	1244	120	start	M05
358	T012	J09	0	6250	1244	144	submit	-
358	T012	J09	0	6250	1244	144	start	M06
416	T013	J10	8	6250	4773	1001	submit	-
428	T003	J03	9	6250	10340	587	fail	M08
477	T011	J08	0	6250	1244	120	finish	M05
502	T012	J09	0	6250	1244	144	finish	M06
524	T005	J05	4	1250	466	202	finish	M02
524	T006	J05	4	1250	466	202	finish	M02
527	T007	J05	4	1250	466	205	finish	M02
527	T008	J05	4	1250	466	205	finish	M02
533	T014	J11	0	6250	1244	223	submit	-
533	T014	J11	0	6250	1244	223	start	M05
534	T015	J12	0	6250	1244	228	submit	-
534	T015	J12	0	6250	1244	228	start	M06
592	T016	J13	0	1250	16	19	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
592	T016	J13	0	1250	16	19	start	M03
605	T002	J02	1	1250	1590	309	finish	M02
605	T013	J10	8	6250	4773	1001	start	M02
611	T016	J13	0	1250	16	19	finish	M03
614	T017	J14	8	6250	466	699	submit	-
660	T018	J15	1	937	466	308	submit	-
660	T018	J15	1	937	466	308	start	M03
662	T019	J16	1	1250	610	306	submit	-
662	T020	J16	1	1250	610	305	submit	-
662	T021	J16	1	1250	610	303	submit	-
662	T019	J16	1	1250	610	306	start	M03
665	T022	J17	1	937	549	138	submit	-
665	T023	J17	1	937	549	81	submit	-
665	T024	J17	1	937	549	88	submit	-
665	T025	J17	1	937	549	78	submit	-
665	T026	J17	1	937	549	68	submit	-
665	T027	J17	1	937	549	196	submit	-
665	T028	J17	1	937	549	76	submit	-
665	T029	J17	1	937	549	76	submit	-
665	T030	J17	1	937	549	88	submit	-
665	T031	J17	1	937	549	106	submit	-
665	T032	J17	1	937	549	88	submit	-
665	T033	J17	1	937	549	90	submit	-
665	T034	J17	1	937	549	77	submit	-
665	T035	J17	1	937	549	81	submit	-
665	T036	J17	1	937	549	111	submit	-
665	T037	J17	1	937	549	72	submit	-
665	T022	J17	1	937	549	138	start	M03
723	T038	J18	1	1874	777	432	submit	-
756	T014	J11	0	6250	1244	223	finish	M05
756	T017	J14	8	6250	466	699	start	M05
762	T015	J12	0	6250	1244	228	finish	M06
762	T020	J16	1	1250	610	305	start	M06
762	T021	J16	1	1250	610	303	start	M06
762	T023	J17	1	937	549	81	start	M06
762	T024	J17	1	937	549	88	start	M06
762	T025	J17	1	937	549	78	start	M06
794	T039	J19	1	1250	1166	341	submit	-
798	T040	J20	0	3125	777	91	submit	-
798	T041	J20	0	3125	777	96	submit	-
798	T042	J20	0	3125	777	117	submit	-
798	T043	J20	0	3125	777	101	submit	-
798	T044	J20	0	3125	777	91	submit	-
798	T045	J20	0	3125	777	104	submit	-
798	T046	J20	0	3125	777	81	submit	-
798	T047	J20	0	3125	777	92	submit	-
798	T048	J20	0	3125	777	104	submit	-
798	T049	J20	0	3125	777	167	submit	-
803	T022	J17	1	937	549	138	finish	M03
803	T026	J17	1	937	549	68	start	M03
819	T050	J21	9	6250	1590	69	submit	-
819	T010	J07	0	6250	1244	468	evict	M01
819	T050	J21	9	6250	1590	69	start	M01
832	T051	J22	0	3125	311	330	submit	-
840	T025	J17	1	937	549	78	finish	M06
840	T027	J17	1	937	549	196	start	M06
843	T023	J17	1	937	549	81	finish	M06

time	task	job	priority	CPU	memory	run time	event	machine
843	T028	J17	1	937	549	76	start	M06
850	T024	J17	1	937	549	88	finish	M06
850	T029	J17	1	937	549	76	start	M06
869	T004	J04	8	6250	4773	569	finish	M04
869	T030	J17	1	937	549	88	start	M04
869	T031	J17	1	937	549	106	start	M04
869	T032	J17	1	937	549	88	start	M04
869	T033	J17	1	937	549	90	start	M04
869	T034	J17	1	937	549	77	start	M04
869	T035	J17	1	937	549	81	start	M04
871	T026	J17	1	937	549	68	finish	M03
871	T036	J17	1	937	549	111	start	M03
888	T050	J21	9	6250	1590	69	finish	M01
888	T037	J17	1	937	549	72	start	M01
888	T038	J18	1	1874	777	432	start	M01
888	T039	J19	1	1250	1166	341	start	M01
892	T052	J23	9	6250	3180	218	submit	-
892	T035	J17	1	937	549	81	evict	M04
892	T034	J17	1	937	549	77	evict	M04
892	T033	J17	1	937	549	90	evict	M04
892	T032	J17	1	937	549	88	evict	M04
892	T031	J17	1	937	549	106	evict	M04
892	T030	J17	1	937	549	88	evict	M04
892	T052	J23	9	6250	3180	218	start	M04
892	T030	J17	1	937	549	88	start	M01
893	T053	J24	0	12500	7959	33	submit	-
893	T054	J24	0	12500	7959	27	submit	-
893	T055	J24	0	12500	7959	31	submit	-
893	T056	J24	0	12500	7959	30	submit	-
893	T057	J24	0	12500	7959	32	submit	-
893	T058	J24	0	12500	7959	31	submit	-
893	T059	J24	0	12500	7959	32	submit	-
893	T060	J24	0	12500	7959	35	submit	-
893	T061	J24	0	12500	7959	31	submit	-
893	T062	J24	0	12500	7959	33	submit	-
893	T063	J24	0	12500	7959	29	submit	-
893	T064	J24	0	12500	7959	28	submit	-
893	T065	J24	0	12500	7959	22	submit	-
893	T066	J24	0	12500	7959	28	submit	-
893	T067	J24	0	12500	7959	36	submit	-
893	T068	J24	0	12500	7959	21	submit	-
894	T069	J25	0	6250	466	43	submit	-
918	T070	J26	4	625	155	34	submit	-
918	T070	J26	4	625	155	34	start	M06
919	T028	J17	1	937	549	76	finish	M06
919	T031	J17	1	937	549	106	start	M06
926	T029	J17	1	937	549	76	finish	M06
926	T032	J17	1	937	549	88	start	M06
952	T070	J26	4	625	155	34	finish	M06
960	T037	J17	1	937	549	72	finish	M01
960	T033	J17	1	937	549	90	start	M01
968	T018	J15	1	937	466	308	finish	M03
968	T019	J16	1	1250	610	306	finish	M03
968	T034	J17	1	937	549	77	start	M03
968	T035	J17	1	937	549	81	start	M03
980	T030	J17	1	937	549	88	finish	M01
982	T036	J17	1	937	549	111	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
986	T071	J27	4	6250	1553	277	submit	-
1013	T072	J28	0	6250	1244	189	submit	-
1014	T032	J17	1	937	549	88	finish	M06
1014	T073	J29	0	6250	1244	190	submit	-
1015	T074	J30	4	1250	466	185	submit	-
1015	T075	J30	4	1250	466	185	submit	-
1015	T076	J30	4	1250	466	184	submit	-
1015	T077	J30	4	1250	466	184	submit	-
1015	T074	J30	4	1250	466	185	start	M01
1015	T075	J30	4	1250	466	185	start	M03
1016	T078	J31	9	6250	3528	3865	submit	-
1016	T076	J30	4	1250	466	184	start	M06
1025	T031	J17	1	937	549	106	finish	M06
1025	T077	J30	4	1250	466	184	start	M06
1036	T027	J17	1	937	549	196	finish	M06
1045	T034	J17	1	937	549	77	finish	M03
1049	T035	J17	1	937	549	81	finish	M03
1050	T033	J17	1	937	549	90	finish	M01
1065	T021	J16	1	1250	610	303	finish	M06
1067	T020	J16	1	1250	610	305	finish	M06
1067	T040	J20	0	3125	777	91	start	M06
1073	T079	J32	1	3125	466	498	submit	-
1075	T080	J33	0	1250	610	390	submit	-
1075	T081	J33	0	1250	610	382	submit	-
1075	T082	J33	0	1250	610	390	submit	-
1075	T083	J33	0	1250	610	399	submit	-
1075	T084	J33	0	1250	610	382	submit	-
1075	T085	J33	0	1250	610	383	submit	-
1075	T086	J33	0	1250	610	401	submit	-
1075	T087	J33	0	1250	610	383	submit	-
1075	T088	J33	0	1250	610	379	submit	-
1075	T089	J33	0	1250	610	393	submit	-
1075	T090	J33	0	1250	610	385	submit	-
1075	T091	J33	0	1250	610	382	submit	-
1075	T092	J33	0	1250	610	389	submit	-
1075	T093	J33	0	1250	610	403	submit	-
1075	T094	J33	0	1250	610	404	submit	-
1075	T095	J33	0	1250	610	412	submit	-
1075	T080	J33	0	1250	610	390	start	M01
1075	T081	J33	0	1250	610	382	start	M03
1087	T096	J34	4	6250	3107	127	submit	-
1087	T097	J34	4	6250	3107	114	submit	-
1087	T098	J34	4	6250	3107	110	submit	-
1087	T099	J34	4	6250	3107	114	submit	-
1110	T052	J23	9	6250	3180	218	finish	M04
1110	T078	J31	9	6250	3528	3865	start	M04
1158	T040	J20	0	3125	777	91	finish	M06
1175	T100	J35	1	1874	777	246	submit	-
1194	T101	J36	9	6250	10340	1063	submit	-
1195	T102	J37	9	6250	3180	218	submit	-
1195	T079	J32	1	3125	466	498	start	M06
1200	T074	J30	4	1250	466	185	finish	M01
1200	T075	J30	4	1250	466	185	finish	M03
1200	T076	J30	4	1250	466	184	finish	M06
1200	T100	J35	1	1874	777	246	start	M03
1200	T082	J33	0	1250	610	390	start	M06

Cell window 04: Events

Time 428: machine M08 failed; task T003 stopped and went back to the queue.

Time 819: task T050 (priority 9) evicted task T010 (priority 0) from machine M01.

Time 892: task T052 (priority 9) evicted task T030 (priority 1) from machine M04.

Time 892: task T052 (priority 9) evicted task T031 (priority 1) from machine M04.

Time 892: task T052 (priority 9) evicted task T032 (priority 1) from machine M04.

Time 892: task T052 (priority 9) evicted task T033 (priority 1) from machine M04.

Time 892: task T052 (priority 9) evicted task T034 (priority 1) from machine M04.

Time 892: task T052 (priority 9) evicted task T035 (priority 1) from machine M04.

Time 892: task T030 restarted on machine M01.

Time 919: task T031 restarted on machine M06.

Time 926: task T032 restarted on machine M06.

Time 960: task T033 restarted on machine M01.

Time 968: task T034 restarted on machine M03.

Time 968: task T035 restarted on machine M03.

Longest waits (submission to first start, tasks started by the window end): task T040 waited 269 s (submitted at 798, first started at 1067); task T037 waited 223 s (submitted at 665, first started at 888); task T036 waited 206 s (submitted at 665, first started at 871); task T030 waited 204 s (submitted at 665, first started at 869); task T031 waited 204 s (submitted at 665, first started at 869).

Cell window 04: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T038	J18	1	1874	777	M01	888	120
T039	J19	1	1250	1166	M01	888	29
T080	J33	0	1250	610	M01	1075	265
T013	J10	8	6250	4773	M02	605	406
T081	J33	0	1250	610	M03	1075	257
T100	J35	1	1874	777	M03	1200	246
T078	J31	9	6250	3528	M04	1110	3775
T017	J14	8	6250	466	M05	756	255
T077	J30	4	1250	466	M06	1025	9
T079	J32	1	3125	466	M06	1195	493
T082	J33	0	1250	610	M06	1200	390
T009	J06	8	6250	4773	M07	341	226

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T003	J03	9	6250	10340	297	587
T101	J36	9	6250	10340	1194	1063
T102	J37	9	6250	3180	1195	218
T071	J27	4	6250	1553	986	277
T096	J34	4	6250	3107	1087	127
T097	J34	4	6250	3107	1087	114
T098	J34	4	6250	3107	1087	110
T099	J34	4	6250	3107	1087	114

task	job	priority	CPU	memory	submitted	run time
T010	J07	0	6250	1244	356	468
T041	J20	0	3125	777	798	96
T042	J20	0	3125	777	798	117
T043	J20	0	3125	777	798	101
T044	J20	0	3125	777	798	91
T045	J20	0	3125	777	798	104
T046	J20	0	3125	777	798	81
T047	J20	0	3125	777	798	92
T048	J20	0	3125	777	798	104
T049	J20	0	3125	777	798	167
T051	J22	0	3125	311	832	330
T053	J24	0	12500	7959	893	33
T054	J24	0	12500	7959	893	27
T055	J24	0	12500	7959	893	31
T056	J24	0	12500	7959	893	30
T057	J24	0	12500	7959	893	32
T058	J24	0	12500	7959	893	31
T059	J24	0	12500	7959	893	32
T060	J24	0	12500	7959	893	35
T061	J24	0	12500	7959	893	31
T062	J24	0	12500	7959	893	33
T063	J24	0	12500	7959	893	29
T064	J24	0	12500	7959	893	28
T065	J24	0	12500	7959	893	22
T066	J24	0	12500	7959	893	28
T067	J24	0	12500	7959	893	36
T068	J24	0	12500	7959	893	21
T069	J25	0	6250	466	894	43
T072	J28	0	6250	1244	1013	189
T073	J29	0	6250	1244	1014	190
T083	J33	0	1250	610	1075	399
T084	J33	0	1250	610	1075	382
T085	J33	0	1250	610	1075	383
T086	J33	0	1250	610	1075	401
T087	J33	0	1250	610	1075	383
T088	J33	0	1250	610	1075	379
T089	J33	0	1250	610	1075	393
T090	J33	0	1250	610	1075	385
T091	J33	0	1250	610	1075	382
T092	J33	0	1250	610	1075	389
T093	J33	0	1250	610	1075	403
T094	J33	0	1250	610	1075	404
T095	J33	0	1250	610	1075	412

Machine M08 is down at time 1200; it does not return.

Cell window 07

Times 0 to 1200. The cluster has 8 machines. 95 tasks of 28 jobs are submitted during the window.

Cell window 07: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	563	1563
M02	6250	6244	-	-
M03	6250	9363	-	-
M04	6250	3116	-	-
M05	6250	3116	-	-
M06	6250	3116	-	-
M07	6250	6244	-	-
M08	6250	9363	-	-

Cell window 07: Task log

time	task	job	priority	CPU	memory	run time	event	machine
74	T001	J01	4	6250	3979	449	submit	-
74	T002	J01	4	6250	3979	448	submit	-
74	T003	J01	4	6250	3979	449	submit	-
74	T004	J01	4	6250	3979	448	submit	-
74	T005	J02	8	6250	1590	150	submit	-
74	T005	J02	8	6250	1590	150	start	M01
74	T001	J01	4	6250	3979	449	start	M02
74	T002	J01	4	6250	3979	448	start	M03
74	T003	J01	4	6250	3979	449	start	M07
74	T004	J01	4	6250	3979	448	start	M08
107	T006	J03	4	3125	1125	47	submit	-
107	T006	J03	4	3125	1125	47	start	M04
139	T007	J04	1	1250	1553	254	submit	-
139	T008	J05	9	4688	6360	783	submit	-
139	T007	J04	1	1250	1553	254	start	M04
154	T006	J03	4	3125	1125	47	finish	M04
191	T009	J06	8	3125	3180	210	submit	-
224	T005	J02	8	6250	1590	150	finish	M01
224	T009	J06	8	3125	3180	210	start	M01
234	T010	J07	0	625	155	287	submit	-
234	T011	J07	0	625	155	250	submit	-
234	T010	J07	0	625	155	287	start	M01
234	T011	J07	0	625	155	250	start	M01
237	T012	J08	0	625	155	275	submit	-
237	T013	J08	0	625	155	278	submit	-
237	T014	J08	0	625	155	280	submit	-
237	T015	J08	0	625	155	281	submit	-
237	T016	J08	0	625	155	274	submit	-
237	T017	J08	0	625	155	279	submit	-
237	T018	J08	0	625	155	306	submit	-
237	T019	J08	0	625	155	277	submit	-
237	T020	J08	0	625	155	291	submit	-
237	T021	J08	0	625	155	282	submit	-
237	T022	J08	0	625	155	304	submit	-
237	T023	J08	0	625	155	300	submit	-
237	T024	J08	0	625	155	279	submit	-
237	T025	J08	0	625	155	283	submit	-
237	T026	J08	0	625	155	280	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
237	T027	J08	0	625	155	275	submit	-
237	T012	J08	0	625	155	275	start	M01
237	T013	J08	0	625	155	278	start	M01
237	T014	J08	0	625	155	280	start	M01
237	T015	J08	0	625	155	281	start	M04
237	T016	J08	0	625	155	274	start	M04
237	T017	J08	0	625	155	279	start	M04
237	T018	J08	0	625	155	306	start	M04
237	T019	J08	0	625	155	277	start	M04
237	T020	J08	0	625	155	291	start	M04
237	T021	J08	0	625	155	282	start	M04
237	T022	J08	0	625	155	304	start	M04
237	T023	J08	0	625	155	300	start	M05
237	T024	J08	0	625	155	279	start	M05
237	T025	J08	0	625	155	283	start	M05
237	T026	J08	0	625	155	280	start	M05
237	T027	J08	0	625	155	275	start	M05
246	T028	J09	0	625	155	294	submit	-
246	T029	J09	0	625	155	314	submit	-
246	T030	J09	0	625	155	296	submit	-
246	T031	J09	0	625	155	306	submit	-
246	T032	J09	0	625	155	304	submit	-
246	T033	J09	0	625	155	306	submit	-
246	T034	J09	0	625	155	309	submit	-
246	T035	J09	0	625	155	285	submit	-
246	T036	J09	0	625	155	289	submit	-
246	T037	J09	0	625	155	290	submit	-
246	T038	J09	0	625	155	296	submit	-
246	T039	J09	0	625	155	298	submit	-
246	T040	J09	0	625	155	288	submit	-
246	T041	J09	0	625	155	288	submit	-
246	T042	J09	0	625	155	287	submit	-
246	T043	J09	0	625	155	286	submit	-
246	T028	J09	0	625	155	294	start	M05
246	T029	J09	0	625	155	314	start	M05
246	T030	J09	0	625	155	296	start	M05
246	T031	J09	0	625	155	306	start	M05
246	T032	J09	0	625	155	304	start	M05
246	T033	J09	0	625	155	306	start	M06
246	T034	J09	0	625	155	309	start	M06
246	T035	J09	0	625	155	285	start	M06
246	T036	J09	0	625	155	289	start	M06
246	T037	J09	0	625	155	290	start	M06
246	T038	J09	0	625	155	296	start	M06
246	T039	J09	0	625	155	298	start	M06
246	T040	J09	0	625	155	288	start	M06
246	T041	J09	0	625	155	288	start	M06
246	T042	J09	0	625	155	287	start	M06
250	T044	J10	1	937	777	61	submit	-
322	T045	J11	4	625	197	66	submit	-
379	T046	J12	0	6250	466	41	submit	-
380	T047	J13	0	6250	1244	78	submit	-
393	T007	J04	1	1250	1553	254	finish	M04
393	T045	J11	4	625	197	66	start	M04
393	T043	J09	0	625	155	286	start	M04
402	T048	J14	4	625	311	329	submit	-
402	T049	J15	4	3125	1125	29	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
434	T009	J06	8	3125	3180	210	finish	M01
434	T048	J14	4	625	311	329	start	M01
434	T044	J10	1	937	777	61	start	M01
447	T050	J16	0	625	219	261	submit	-
447	T051	J16	0	625	219	260	submit	-
447	T052	J16	0	625	219	258	submit	-
447	T053	J16	0	625	219	258	submit	-
447	T054	J16	0	625	219	259	submit	-
447	T055	J16	0	625	219	258	submit	-
447	T056	J16	0	625	219	257	submit	-
447	T057	J16	0	625	219	257	submit	-
447	T058	J16	0	625	219	257	submit	-
447	T059	J16	0	625	219	257	submit	-
447	T060	J16	0	625	219	260	submit	-
447	T061	J16	0	625	219	257	submit	-
447	T062	J16	0	625	219	260	submit	-
447	T063	J16	0	625	219	259	submit	-
447	T064	J16	0	625	219	259	submit	-
447	T065	J16	0	625	219	261	submit	-
447	T050	J16	0	625	219	261	start	M01
447	T051	J16	0	625	219	260	start	M01
459	T045	J11	4	625	197	66	finish	M04
459	T052	J16	0	625	219	258	start	M04
484	T011	J07	0	625	155	250	finish	M01
484	T053	J16	0	625	219	258	start	M01
495	T044	J10	1	937	777	61	finish	M01
495	T054	J16	0	625	219	259	start	M01
495	T055	J16	0	625	219	258	start	M01
498	T066	J17	8	5090	466	824	submit	-
511	T016	J08	0	625	155	274	finish	M04
511	T056	J16	0	625	219	257	start	M04
512	T012	J08	0	625	155	275	finish	M01
512	T027	J08	0	625	155	275	finish	M05
512	T057	J16	0	625	219	257	start	M01
512	T058	J16	0	625	219	257	start	M05
514	T019	J08	0	625	155	277	finish	M04
514	T059	J16	0	625	219	257	start	M04
515	T013	J08	0	625	155	278	finish	M01
515	T067	J18	4	1250	610	779	submit	-
515	T068	J18	4	1250	610	737	submit	-
515	T069	J18	4	1250	610	650	submit	-
515	T070	J18	4	1250	610	640	submit	-
515	T060	J16	0	625	219	260	start	M01
516	T017	J08	0	625	155	279	finish	M04
516	T024	J08	0	625	155	279	finish	M05
516	T061	J16	0	625	219	257	start	M04
516	T062	J16	0	625	219	260	start	M05
517	T014	J08	0	625	155	280	finish	M01
517	T026	J08	0	625	155	280	finish	M05
517	T063	J16	0	625	219	259	start	M01
517	T064	J16	0	625	219	259	start	M05
518	T015	J08	0	625	155	281	finish	M04
518	T065	J16	0	625	219	261	start	M04
519	T021	J08	0	625	155	282	finish	M04
520	T025	J08	0	625	155	283	finish	M05
521	T010	J07	0	625	155	287	finish	M01
522	T002	J01	4	6250	3979	448	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
522	T004	J01	4	6250	3979	448	finish	M08
522	T008	J05	9	4688	6360	783	start	M03
522	T066	J17	8	5090	466	824	start	M08
522	T067	J18	4	1250	610	779	start	M03
523	T001	J01	4	6250	3979	449	finish	M02
523	T003	J01	4	6250	3979	449	finish	M07
523	T049	J15	4	3125	1125	29	start	M02
523	T068	J18	4	1250	610	737	start	M02
523	T069	J18	4	1250	610	650	start	M02
523	T070	J18	4	1250	610	640	start	M07
528	T020	J08	0	625	155	291	finish	M04
531	T035	J09	0	625	155	285	finish	M06
533	T042	J09	0	625	155	287	finish	M06
534	T040	J09	0	625	155	288	finish	M06
534	T041	J09	0	625	155	288	finish	M06
535	T036	J09	0	625	155	289	finish	M06
536	T037	J09	0	625	155	290	finish	M06
537	T023	J08	0	625	155	300	finish	M05
540	T028	J09	0	625	155	294	finish	M05
541	T022	J08	0	625	155	304	finish	M04
542	T030	J09	0	625	155	296	finish	M05
542	T038	J09	0	625	155	296	finish	M06
543	T018	J08	0	625	155	306	finish	M04
544	T039	J09	0	625	155	298	finish	M06
550	T032	J09	0	625	155	304	finish	M05
552	T031	J09	0	625	155	306	finish	M05
552	T033	J09	0	625	155	306	finish	M06
552	T049	J15	4	3125	1125	29	finish	M02
555	T034	J09	0	625	155	309	finish	M06
555	T046	J12	0	6250	466	41	start	M06
560	T029	J09	0	625	155	314	finish	M05
563	T048	J14	4	625	311	329	fail	M01
563	T050	J16	0	625	219	261	fail	M01
563	T051	J16	0	625	219	260	fail	M01
563	T053	J16	0	625	219	258	fail	M01
563	T054	J16	0	625	219	259	fail	M01
563	T055	J16	0	625	219	258	fail	M01
563	T057	J16	0	625	219	257	fail	M01
563	T060	J16	0	625	219	260	fail	M01
563	T063	J16	0	625	219	259	fail	M01
563	T048	J14	4	625	311	329	start	M02
563	T050	J16	0	625	219	261	start	M02
563	T051	J16	0	625	219	260	start	M02
563	T053	J16	0	625	219	258	start	M02
563	T054	J16	0	625	219	259	start	M02
563	T055	J16	0	625	219	258	start	M02
563	T057	J16	0	625	219	257	start	M04
563	T060	J16	0	625	219	260	start	M04
563	T063	J16	0	625	219	259	start	M04
596	T046	J12	0	6250	466	41	finish	M06
596	T047	J13	0	6250	1244	78	start	M06
674	T047	J13	0	6250	1244	78	finish	M06
679	T043	J09	0	625	155	286	finish	M04
679	T071	J19	9	6250	3180	330	submit	-
717	T052	J16	0	625	219	258	finish	M04
768	T056	J16	0	625	219	257	finish	M04
769	T058	J16	0	625	219	257	finish	M05

time	task	job	priority	CPU	memory	run time	event	machine
771	T059	J16	0	625	219	257	finish	M04
773	T061	J16	0	625	219	257	finish	M04
776	T062	J16	0	625	219	260	finish	M05
776	T064	J16	0	625	219	259	finish	M05
779	T065	J16	0	625	219	261	finish	M04
820	T057	J16	0	625	219	257	finish	M04
821	T053	J16	0	625	219	258	finish	M02
821	T055	J16	0	625	219	258	finish	M02
822	T054	J16	0	625	219	259	finish	M02
822	T063	J16	0	625	219	259	finish	M04
823	T051	J16	0	625	219	260	finish	M02
823	T060	J16	0	625	219	260	finish	M04
824	T050	J16	0	625	219	261	finish	M02
859	T072	J20	9	62	2	26	submit	-
859	T072	J20	9	62	2	26	start	M02
871	T073	J21	0	3125	311	2953	submit	-
871	T074	J21	0	3125	311	2967	submit	-
871	T075	J21	0	3125	311	2986	submit	-
871	T076	J21	0	3125	311	2978	submit	-
871	T077	J21	0	3125	311	2995	submit	-
871	T078	J21	0	3125	311	2962	submit	-
871	T079	J21	0	3125	311	3026	submit	-
871	T080	J21	0	3125	311	2954	submit	-
871	T081	J21	0	3125	311	3013	submit	-
871	T082	J21	0	3125	311	2952	submit	-
871	T083	J21	0	3125	311	2998	submit	-
871	T084	J21	0	3125	311	3006	submit	-
871	T085	J21	0	3125	311	2976	submit	-
871	T086	J21	0	3125	311	3049	submit	-
871	T087	J21	0	3125	311	2966	submit	-
871	T088	J21	0	3125	311	2978	submit	-
871	T073	J21	0	3125	311	2953	start	M04
871	T074	J21	0	3125	311	2967	start	M04
871	T075	J21	0	3125	311	2986	start	M05
871	T076	J21	0	3125	311	2978	start	M05
871	T077	J21	0	3125	311	2995	start	M06
871	T078	J21	0	3125	311	2962	start	M06
885	T072	J20	9	62	2	26	finish	M02
885	T079	J21	0	3125	311	3026	start	M02
892	T048	J14	4	625	311	329	finish	M02
919	T089	J22	4	3125	1125	60	submit	-
919	T090	J23	4	3125	1224	142	submit	-
919	T091	J24	4	3125	1125	35	submit	-
919	T089	J22	4	3125	1125	60	start	M07
924	T092	J25	9	6250	1590	92	submit	-
979	T089	J22	4	3125	1125	60	finish	M07
979	T090	J23	4	3125	1224	142	start	M07
980	T093	J26	0	6250	466	44	submit	-
990	T094	J27	4	1250	398	32	submit	-
990	T094	J27	4	1250	398	32	start	M07
1022	T094	J27	4	1250	398	32	finish	M07
1101	T095	J28	9	4688	6360	579	submit	-
1121	T090	J23	4	3125	1224	142	finish	M07
1121	T091	J24	4	3125	1125	35	start	M07
1156	T091	J24	4	3125	1125	35	finish	M07
1163	T070	J18	4	1250	610	640	finish	M07
1163	T071	J19	9	6250	3180	330	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
1163	T074	J21	0	3125	311	2967	evict	M04
1163	T073	J21	0	3125	311	2953	evict	M04
1163	T092	J25	9	6250	1590	92	start	M04
1173	T069	J18	4	1250	610	650	finish	M02

Cell window 07: Events

Time 563: machine M01 failed; tasks T048, T050, T051, T053, T054, T055, T057, T060, T063 stopped and went back to the queue.

Time 563: task T048 restarted on machine M02.

Time 563: task T050 restarted on machine M02.

Time 563: task T051 restarted on machine M02.

Time 563: task T053 restarted on machine M02.

Time 563: task T054 restarted on machine M02.

Time 563: task T055 restarted on machine M02.

Time 563: task T057 restarted on machine M04.

Time 563: task T060 restarted on machine M04.

Time 563: task T063 restarted on machine M04.

Time 1163: task T092 (priority 9) evicted task T073 (priority 0) from machine M04.

Time 1163: task T092 (priority 9) evicted task T074 (priority 0) from machine M04.

Longest waits (submission to first start, tasks started by the window end): task T071 waited 484 s (submitted at 679, first started at 1163); task T008 waited 383 s (submitted at 139, first started at 522); task T092 waited 239 s (submitted at 924, first started at 1163); task T047 waited 216 s (submitted at 380, first started at 596); task T091 waited 202 s (submitted at 919, first started at 1121).

Cell window 07: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T068	J18	4	1250	610	M02	523	60
T079	J21	0	3125	311	M02	885	2711
T008	J05	9	4688	6360	M03	522	105
T067	J18	4	1250	610	M03	522	101
T092	J25	9	6250	1590	M04	1163	55
T075	J21	0	3125	311	M05	871	2657
T076	J21	0	3125	311	M05	871	2649
T077	J21	0	3125	311	M06	871	2666
T078	J21	0	3125	311	M06	871	2633
T071	J19	9	6250	3180	M07	1163	293
T066	J17	8	5090	466	M08	522	146

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T095	J28	9	4688	6360	1101	579
T073	J21	0	3125	311	871	2953
T074	J21	0	3125	311	871	2967
T080	J21	0	3125	311	871	2954
T081	J21	0	3125	311	871	3013

task	job	priority	CPU	memory	submitted	run time
T082	J21	0	3125	311	871	2952
T083	J21	0	3125	311	871	2998
T084	J21	0	3125	311	871	3006
T085	J21	0	3125	311	871	2976
T086	J21	0	3125	311	871	3049
T087	J21	0	3125	311	871	2966
T088	J21	0	3125	311	871	2978
T093	J26	0	6250	466	980	44

Machine M01 is down at time 1200; it returns at time 1563.

Cell window 11

Times 0 to 1200. The cluster has 8 machines. 136 tasks of 63 jobs are submitted during the window.

Cell window 11: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	-	-
M02	6250	6244	-	-
M03	6250	6244	-	-
M04	6250	3116	-	-
M05	6250	6244	-	-
M06	6250	3116	318	1321
M07	6250	6244	-	-
M08	6250	6244	-	-

Cell window 11: Task log

time	task	job	priority	CPU	memory	run time	event	machine
2	T001	J01	1	1562	1553	365	submit	-
2	T001	J01	1	1562	1553	365	start	M01
8	T002	J02	8	1250	777	70	submit	-
8	T002	J02	8	1250	777	70	start	M01
35	T003	J03	1	1639	1553	313	submit	-
35	T003	J03	1	1639	1553	313	start	M01
39	T004	J04	4	1250	466	266	submit	-
39	T005	J04	4	1250	466	284	submit	-
39	T006	J04	4	1250	466	224	submit	-
39	T007	J04	4	1250	466	222	submit	-
39	T008	J05	0	1250	835	148	submit	-
39	T009	J05	0	1250	835	200	submit	-
39	T010	J05	0	1250	835	218	submit	-
39	T011	J05	0	1250	835	167	submit	-
39	T012	J05	0	1250	835	143	submit	-
39	T013	J05	0	1250	835	175	submit	-
39	T014	J05	0	1250	835	196	submit	-
39	T015	J05	0	1250	835	136	submit	-
39	T016	J05	0	1250	835	199	submit	-
39	T017	J05	0	1250	835	125	submit	-
39	T018	J05	0	1250	835	147	submit	-
39	T019	J05	0	1250	835	217	submit	-
39	T020	J05	0	1250	835	308	submit	-
39	T021	J05	0	1250	835	234	submit	-
39	T022	J05	0	1250	835	144	submit	-
39	T023	J05	0	1250	835	306	submit	-
39	T004	J04	4	1250	466	266	start	M01
39	T005	J04	4	1250	466	284	start	M02
39	T006	J04	4	1250	466	224	start	M02
39	T007	J04	4	1250	466	222	start	M02
39	T008	J05	0	1250	835	148	start	M02
39	T009	J05	0	1250	835	200	start	M02
39	T010	J05	0	1250	835	218	start	M03
39	T011	J05	0	1250	835	167	start	M03
39	T012	J05	0	1250	835	143	start	M03
39	T013	J05	0	1250	835	175	start	M03
39	T014	J05	0	1250	835	196	start	M03
39	T015	J05	0	1250	835	136	start	M04

time	task	job	priority	CPU	memory	run time	event	machine
39	T016	J05	0	1250	835	199	start	M04
39	T017	J05	0	1250	835	125	start	M04
39	T018	J05	0	1250	835	147	start	M05
39	T019	J05	0	1250	835	217	start	M05
39	T020	J05	0	1250	835	308	start	M05
39	T021	J05	0	1250	835	234	start	M05
39	T022	J05	0	1250	835	144	start	M05
39	T023	J05	0	1250	835	306	start	M06
42	T024	J06	1	1562	1553	105	submit	-
42	T024	J06	1	1562	1553	105	start	M06
50	T025	J07	8	1250	1166	460	submit	-
50	T025	J07	8	1250	1166	460	start	M07
58	T026	J08	0	6250	466	44	submit	-
58	T026	J08	0	6250	466	44	start	M08
62	T027	J09	1	1250	1590	432	submit	-
62	T027	J09	1	1250	1590	432	start	M07
66	T028	J10	1	3125	2386	245	submit	-
66	T028	J10	1	3125	2386	245	start	M07
68	T029	J11	4	625	311	240	submit	-
68	T029	J11	4	625	311	240	start	M04
78	T002	J02	8	1250	777	70	finish	M01
83	T030	J12	1	1250	1553	53	submit	-
83	T030	J12	1	1250	1553	53	start	M01
85	T031	J13	0	1250	197	49	submit	-
85	T031	J13	0	1250	197	49	start	M04
90	T032	J14	1	1250	1553	46	submit	-
102	T026	J08	0	6250	466	44	finish	M08
102	T032	J14	1	1250	1553	46	start	M08
127	T033	J15	1	1562	1553	116	submit	-
127	T033	J15	1	1562	1553	116	start	M08
134	T031	J13	0	1250	197	49	finish	M04
136	T030	J12	1	1250	1553	53	finish	M01
147	T024	J06	1	1562	1553	105	finish	M06
147	T034	J16	8	625	155	97	submit	-
147	T034	J16	8	625	155	97	start	M01
148	T032	J14	1	1250	1553	46	finish	M08
164	T017	J05	0	1250	835	125	finish	M04
165	T035	J17	6	937	1244	84	submit	-
165	T035	J17	6	937	1244	84	start	M01
175	T015	J05	0	1250	835	136	finish	M04
182	T012	J05	0	1250	835	143	finish	M03
183	T022	J05	0	1250	835	144	finish	M05
186	T018	J05	0	1250	835	147	finish	M05
187	T008	J05	0	1250	835	148	finish	M02
203	T036	J18	6	937	280	126	submit	-
203	T037	J18	6	937	280	59	submit	-
203	T038	J18	6	937	280	76	submit	-
203	T039	J18	6	937	280	90	submit	-
203	T036	J18	6	937	280	126	start	M02
203	T037	J18	6	937	280	59	start	M03
203	T038	J18	6	937	280	76	start	M04
203	T039	J18	6	937	280	90	start	M04
206	T011	J05	0	1250	835	167	finish	M03
207	T040	J19	1	1562	1553	325	submit	-
207	T040	J19	1	1562	1553	325	start	M03
214	T013	J05	0	1250	835	175	finish	M03
221	T041	J20	1	1562	1553	97	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
221	T041	J20	1	1562	1553	97	start	M05
228	T042	J21	1	1562	1553	301	submit	-
228	T042	J21	1	1562	1553	301	start	M06
235	T014	J05	0	1250	835	196	finish	M03
238	T016	J05	0	1250	835	199	finish	M04
239	T009	J05	0	1250	835	200	finish	M02
243	T033	J15	1	1562	1553	116	finish	M08
244	T034	J16	8	625	155	97	finish	M01
249	T035	J17	6	937	1244	84	finish	M01
256	T019	J05	0	1250	835	217	finish	M05
257	T010	J05	0	1250	835	218	finish	M03
259	T043	J22	1	1250	1553	256	submit	-
259	T043	J22	1	1250	1553	256	start	M01
261	T007	J04	4	1250	466	222	finish	M02
261	T044	J23	0	905	227	254	submit	-
261	T044	J23	0	905	227	254	start	M02
262	T037	J18	6	937	280	59	finish	M03
263	T006	J04	4	1250	466	224	finish	M02
263	T045	J24	1	1250	1553	254	submit	-
263	T045	J24	1	1250	1553	254	start	M02
265	T046	J25	0	1250	283	252	submit	-
265	T046	J25	0	1250	283	252	start	M02
273	T021	J05	0	1250	835	234	finish	M05
279	T038	J18	6	937	280	76	finish	M04
293	T039	J18	6	937	280	90	finish	M04
305	T004	J04	4	1250	466	266	finish	M01
308	T029	J11	4	625	311	240	finish	M04
308	T047	J26	1	1562	1553	227	submit	-
308	T047	J26	1	1562	1553	227	start	M01
311	T028	J10	1	3125	2386	245	finish	M07
318	T041	J20	1	1562	1553	97	finish	M05
318	T023	J05	0	1250	835	306	fail	M06
318	T042	J21	1	1562	1553	301	fail	M06
318	T042	J21	1	1562	1553	301	start	M03
318	T023	J05	0	1250	835	306	start	M03
323	T005	J04	4	1250	466	284	finish	M02
329	T036	J18	6	937	280	126	finish	M02
329	T048	J27	8	625	1553	1276	submit	-
329	T048	J27	8	625	1553	1276	start	M02
347	T020	J05	0	1250	835	308	finish	M05
348	T003	J03	1	1639	1553	313	finish	M01
355	T049	J28	1	1250	1553	169	submit	-
355	T050	J29	0	1250	16	27	submit	-
355	T049	J28	1	1250	1553	169	start	M01
355	T050	J29	0	1250	16	27	start	M02
356	T051	J30	0	1250	204	167	submit	-
356	T051	J30	0	1250	204	167	start	M03
358	T052	J31	0	1250	197	152	submit	-
358	T053	J31	0	1250	197	106	submit	-
358	T054	J31	0	1250	197	127	submit	-
358	T052	J31	0	1250	197	152	start	M04
358	T053	J31	0	1250	197	106	start	M04
358	T054	J31	0	1250	197	127	start	M04
367	T001	J01	1	1562	1553	365	finish	M01
373	T055	J32	4	1250	466	334	submit	-
373	T056	J33	8	6250	466	1213	submit	-
373	T056	J33	8	6250	466	1213	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
373	T055	J32	4	1250	466	334	start	M01
375	T057	J34	4	722	197	314	submit	-
375	T057	J34	4	722	197	314	start	M01
382	T050	J29	0	1250	16	27	finish	M02
391	T058	J35	4	6250	622	463	submit	-
391	T059	J35	4	6250	622	423	submit	-
391	T060	J35	4	6250	622	430	submit	-
391	T061	J35	4	6250	622	364	submit	-
391	T058	J35	4	6250	622	463	start	M08
396	T062	J36	9	6250	1590	86	submit	-
396	T051	J30	0	1250	204	167	evict	M03
396	T023	J05	0	1250	835	306	evict	M03
396	T042	J21	1	1562	1553	301	evict	M03
396	T040	J19	1	1562	1553	325	evict	M03
396	T062	J36	9	6250	1590	86	start	M03
396	T040	J19	1	1562	1553	325	start	M02
396	T042	J21	1	1562	1553	301	start	M04
396	T023	J05	0	1250	835	306	start	M07
396	T051	J30	0	1250	204	167	start	M07
401	T063	J37	1	1562	1553	247	submit	-
406	T064	J38	1	937	466	1137	submit	-
406	T064	J38	1	937	466	1137	start	M04
409	T065	J39	0	1250	610	1134	submit	-
409	T066	J39	0	1250	610	1124	submit	-
409	T067	J39	0	1250	610	1125	submit	-
409	T068	J39	0	1250	610	1122	submit	-
409	T069	J39	0	1250	610	1122	submit	-
409	T070	J39	0	1250	610	1121	submit	-
409	T071	J39	0	1250	610	1122	submit	-
409	T072	J39	0	1250	610	1123	submit	-
409	T073	J39	0	1250	610	1122	submit	-
409	T074	J39	0	1250	610	1121	submit	-
409	T075	J39	0	1250	610	1123	submit	-
409	T076	J39	0	1250	610	81	submit	-
409	T065	J39	0	1250	610	1134	start	M07
446	T077	J40	1	1250	1553	557	submit	-
448	T078	J41	1	1250	1553	182	submit	-
450	T079	J42	0	1250	932	164	submit	-
450	T080	J42	0	1250	932	153	submit	-
450	T081	J42	0	1250	932	191	submit	-
450	T082	J42	0	1250	932	209	submit	-
450	T083	J42	0	1250	932	191	submit	-
450	T084	J42	0	1250	932	149	submit	-
450	T085	J42	0	1250	932	552	submit	-
450	T086	J42	0	1250	932	194	submit	-
450	T087	J42	0	1250	932	152	submit	-
450	T088	J42	0	1250	932	149	submit	-
450	T089	J42	0	1250	932	152	submit	-
450	T090	J42	0	1250	932	153	submit	-
450	T091	J42	0	1250	932	150	submit	-
450	T092	J42	0	1250	932	173	submit	-
450	T093	J42	0	1250	932	185	submit	-
450	T094	J42	0	1250	932	163	submit	-
464	T053	J31	0	1250	197	106	finish	M04
464	T066	J39	0	1250	610	1124	start	M04
482	T062	J36	9	6250	1590	86	finish	M03
482	T059	J35	4	6250	622	423	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
485	T054	J31	0	1250	197	127	finish	M04
494	T027	J09	1	1250	1590	432	finish	M07
494	T077	J40	1	1250	1553	557	start	M07
510	T025	J07	8	1250	1166	460	finish	M07
510	T052	J31	0	1250	197	152	finish	M04
510	T078	J41	1	1250	1553	182	start	M07
515	T043	J22	1	1250	1553	256	finish	M01
515	T044	J23	0	905	227	254	finish	M02
515	T067	J39	0	1250	610	1125	start	M02
515	T076	J39	0	1250	610	81	start	M01
517	T045	J24	1	1250	1553	254	finish	M02
517	T046	J25	0	1250	283	252	finish	M02
517	T063	J37	1	1562	1553	247	start	M02
517	T068	J39	0	1250	610	1122	start	M02
524	T049	J28	1	1250	1553	169	finish	M01
524	T079	J42	0	1250	932	164	start	M01
529	T095	J43	4	1250	466	287	submit	-
529	T095	J43	4	1250	466	287	start	M04
531	T096	J44	4	1250	466	263	submit	-
531	T097	J44	4	1250	466	263	submit	-
531	T098	J44	4	1250	466	265	submit	-
531	T099	J44	4	1250	466	268	submit	-
535	T047	J26	1	1562	1553	227	finish	M01
535	T080	J42	0	1250	932	153	start	M01
546	T100	J45	1	1562	1553	121	submit	-
563	T051	J30	0	1250	204	167	finish	M07
563	T096	J44	4	1250	466	263	start	M07
565	T101	J46	1	1562	1553	249	submit	-
593	T102	J47	1	1250	1553	205	submit	-
593	T103	J48	1	1250	1553	57	submit	-
595	T104	J49	0	3125	311	617	submit	-
596	T076	J39	0	1250	610	81	finish	M01
596	T103	J48	1	1250	1553	57	start	M01
651	T105	J50	4	3125	777	238	submit	-
653	T103	J48	1	1250	1553	57	finish	M01
656	T106	J51	9	6250	3180	210	submit	-
688	T079	J42	0	1250	932	164	finish	M01
688	T080	J42	0	1250	932	153	finish	M01
689	T057	J34	4	722	197	314	finish	M01
692	T078	J41	1	1250	1553	182	finish	M07
692	T097	J44	4	1250	466	263	start	M07
697	T042	J21	1	1562	1553	301	finish	M04
697	T098	J44	4	1250	466	265	start	M04
697	T099	J44	4	1250	466	268	start	M04
702	T023	J05	0	1250	835	306	finish	M07
702	T102	J47	1	1250	1553	205	start	M07
707	T055	J32	4	1250	466	334	finish	M01
707	T106	J51	9	6250	3180	210	start	M01
721	T040	J19	1	1562	1553	325	finish	M02
721	T100	J45	1	1562	1553	121	start	M02
764	T063	J37	1	1562	1553	247	finish	M02
764	T101	J46	1	1562	1553	249	start	M02
777	T107	J52	0	6250	1244	112	submit	-
816	T095	J43	4	1250	466	287	finish	M04
816	T069	J39	0	1250	610	1122	start	M04
826	T096	J44	4	1250	466	263	finish	M07
826	T070	J39	0	1250	610	1121	start	M07

time	task	job	priority	CPU	memory	run time	event	machine
842	T100	J45	1	1562	1553	121	finish	M02
842	T071	J39	0	1250	610	1122	start	M02
854	T058	J35	4	6250	622	463	finish	M08
854	T060	J35	4	6250	622	430	start	M08
905	T059	J35	4	6250	622	423	finish	M03
905	T061	J35	4	6250	622	364	start	M03
907	T102	J47	1	1250	1553	205	finish	M07
907	T072	J39	0	1250	610	1123	start	M07
909	T108	J53	4	3125	1125	101	submit	-
917	T106	J51	9	6250	3180	210	finish	M01
917	T105	J50	4	3125	777	238	start	M01
917	T108	J53	4	3125	1125	101	start	M01
955	T097	J44	4	1250	466	263	finish	M07
955	T073	J39	0	1250	610	1122	start	M07
956	T109	J54	1	5627	199	48	submit	-
962	T098	J44	4	1250	466	265	finish	M04
962	T074	J39	0	1250	610	1121	start	M04
965	T099	J44	4	1250	466	268	finish	M04
965	T075	J39	0	1250	610	1123	start	M04
1013	T101	J46	1	1562	1553	249	finish	M02
1013	T081	J42	0	1250	932	191	start	M02
1018	T108	J53	4	3125	1125	101	finish	M01
1051	T077	J40	1	1250	1553	557	finish	M07
1051	T082	J42	0	1250	932	209	start	M07
1077	T110	J55	1	625	233	360	submit	-
1077	T110	J55	1	625	233	360	start	M02
1078	T111	J56	1	625	233	343	submit	-
1117	T112	J57	9	6250	1590	230	submit	-
1117	T075	J39	0	1250	610	1123	evict	M04
1117	T074	J39	0	1250	610	1121	evict	M04
1117	T069	J39	0	1250	610	1122	evict	M04
1117	T066	J39	0	1250	610	1124	evict	M04
1117	T064	J38	1	937	466	1137	evict	M04
1117	T112	J57	9	6250	1590	230	start	M04
1117	T064	J38	1	937	466	1137	start	M01
1117	T111	J56	1	625	233	343	start	M01
1117	T066	J39	0	1250	610	1124	start	M01
1128	T113	J58	1	769	466	276	submit	-
1130	T114	J59	0	625	155	277	submit	-
1130	T115	J59	0	625	155	249	submit	-
1130	T116	J60	0	625	155	287	submit	-
1130	T117	J60	0	625	155	273	submit	-
1130	T118	J61	0	625	155	274	submit	-
1130	T119	J61	0	625	155	269	submit	-
1132	T120	J62	0	625	155	267	submit	-
1132	T121	J62	0	625	155	272	submit	-
1132	T122	J62	0	625	155	268	submit	-
1132	T123	J62	0	625	155	269	submit	-
1132	T124	J62	0	625	155	266	submit	-
1132	T125	J62	0	625	155	271	submit	-
1132	T126	J62	0	625	155	272	submit	-
1132	T127	J62	0	625	155	266	submit	-
1132	T128	J62	0	625	155	267	submit	-
1132	T129	J62	0	625	155	269	submit	-
1132	T130	J62	0	625	155	269	submit	-
1132	T131	J62	0	625	155	268	submit	-
1132	T132	J62	0	625	155	272	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1132	T133	J62	0	625	155	268	submit	-
1132	T134	J62	0	625	155	266	submit	-
1132	T135	J62	0	625	155	272	submit	-
1155	T105	J50	4	3125	777	238	finish	M01
1155	T113	J58	1	769	466	276	start	M01
1155	T069	J39	0	1250	610	1122	start	M01
1155	T074	J39	0	1250	610	1121	start	M01
1197	T136	J63	1	625	311	39	submit	-

Cell window 11: Events

Time 318: machine M06 failed; tasks T023, T042 stopped and went back to the queue.

Time 318: task T023 restarted on machine M03.

Time 318: task T042 restarted on machine M03.

Time 396: task T062 (priority 9) evicted task T023 (priority 0) from machine M03.

Time 396: task T062 (priority 9) evicted task T040 (priority 1) from machine M03.

Time 396: task T062 (priority 9) evicted task T042 (priority 1) from machine M03.

Time 396: task T062 (priority 9) evicted task T051 (priority 0) from machine M03.

Time 396: task T023 restarted on machine M07.

Time 396: task T040 restarted on machine M02.

Time 396: task T042 restarted on machine M04.

Time 396: task T051 restarted on machine M07.

Time 1117: task T112 (priority 9) evicted task T064 (priority 1) from machine M04.

Time 1117: task T112 (priority 9) evicted task T066 (priority 0) from machine M04.

Time 1117: task T112 (priority 9) evicted task T069 (priority 0) from machine M04.

Time 1117: task T112 (priority 9) evicted task T074 (priority 0) from machine M04.

Time 1117: task T112 (priority 9) evicted task T075 (priority 0) from machine M04.

Time 1117: task T064 restarted on machine M01.

Time 1117: task T066 restarted on machine M01.

Time 1155: task T069 restarted on machine M01.

Time 1155: task T074 restarted on machine M01.

Longest waits (submission to first start, tasks started by the window end): task T082 waited 601 s (submitted at 450, first started at 1051); task T081 waited 563 s (submitted at 450, first started at 1013); task T075 waited 556 s (submitted at 409, first started at 965); task T074 waited 553 s (submitted at 409, first started at 962); task T073 waited 546 s (submitted at 409, first started at 955).

Cell window 11: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T064	J38	1	937	466	M01	1117	1054
T066	J39	0	1250	610	M01	1117	1041
T069	J39	0	1250	610	M01	1155	1077
T074	J39	0	1250	610	M01	1155	1076

task	job	priority	CPU	memory	machine	started	remaining
T111	J56	1	625	233	M01	1117	260
T113	J58	1	769	466	M01	1155	231
T048	J27	8	625	1553	M02	329	405
T067	J39	0	1250	610	M02	515	440
T068	J39	0	1250	610	M02	517	439
T071	J39	0	1250	610	M02	842	764
T081	J42	0	1250	932	M02	1013	4
T110	J55	1	625	233	M02	1077	237
T061	J35	4	6250	622	M03	905	69
T112	J57	9	6250	1590	M04	1117	147
T056	J33	8	6250	466	M05	373	386
T065	J39	0	1250	610	M07	409	343
T070	J39	0	1250	610	M07	826	747
T072	J39	0	1250	610	M07	907	830
T073	J39	0	1250	610	M07	955	877
T082	J42	0	1250	932	M07	1051	60
T060	J35	4	6250	622	M08	854	84

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T109	J54	1	5627	199	956	48
T136	J63	1	625	311	1197	39
T075	J39	0	1250	610	409	1123
T083	J42	0	1250	932	450	191
T084	J42	0	1250	932	450	149
T085	J42	0	1250	932	450	552
T086	J42	0	1250	932	450	194
T087	J42	0	1250	932	450	152
T088	J42	0	1250	932	450	149
T089	J42	0	1250	932	450	152
T090	J42	0	1250	932	450	153
T091	J42	0	1250	932	450	150
T092	J42	0	1250	932	450	173
T093	J42	0	1250	932	450	185
T094	J42	0	1250	932	450	163
T104	J49	0	3125	311	595	617
T107	J52	0	6250	1244	777	112
T114	J59	0	625	155	1130	277
T115	J59	0	625	155	1130	249
T116	J60	0	625	155	1130	287
T117	J60	0	625	155	1130	273
T118	J61	0	625	155	1130	274
T119	J61	0	625	155	1130	269
T120	J62	0	625	155	1132	267
T121	J62	0	625	155	1132	272
T122	J62	0	625	155	1132	268
T123	J62	0	625	155	1132	269
T124	J62	0	625	155	1132	266
T125	J62	0	625	155	1132	271
T126	J62	0	625	155	1132	272
T127	J62	0	625	155	1132	266
T128	J62	0	625	155	1132	267
T129	J62	0	625	155	1132	269
T130	J62	0	625	155	1132	269
T131	J62	0	625	155	1132	268
T132	J62	0	625	155	1132	272
T133	J62	0	625	155	1132	268

task	job	priority	CPU	memory	submitted	run time
T134	J62	0	625	155	1132	266
T135	J62	0	625	155	1132	272

Machine M06 is down at time 1200; it returns at time 1321.

Cell window 18

Times 0 to 1200. The cluster has 8 machines. 69 tasks of 39 jobs are submitted during the window.

Cell window 18: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	3116	-	-
M03	6250	9363	-	-
M04	6250	6244	-	-
M05	6250	6244	518	never
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	3125	3123	-	-

Cell window 18: Task log

time	task	job	priority	CPU	memory	run time	event	machine
4	T001	J01	8	1250	1166	84	submit	-
4	T001	J01	8	1250	1166	84	start	M01
71	T002	J02	1	4779	2332	5867	submit	-
71	T002	J02	1	4779	2332	5867	start	M02
88	T001	J01	8	1250	1166	84	finish	M01
97	T003	J03	0	125	1553	5684	submit	-
97	T004	J03	0	125	1553	5685	submit	-
97	T005	J03	0	125	1553	5792	submit	-
97	T006	J03	0	125	1553	5685	submit	-
97	T007	J03	0	125	1553	5678	submit	-
97	T008	J03	0	125	1553	5684	submit	-
97	T009	J03	0	125	1553	5705	submit	-
97	T010	J03	0	125	1553	5744	submit	-
97	T011	J03	0	125	1553	5722	submit	-
97	T012	J03	0	125	1553	5664	submit	-
97	T013	J03	0	125	1553	5837	submit	-
97	T014	J03	0	125	1553	5700	submit	-
97	T015	J03	0	125	1553	5681	submit	-
97	T016	J03	0	125	1553	5721	submit	-
97	T017	J03	0	125	1553	5804	submit	-
97	T018	J03	0	125	1553	5705	submit	-
97	T003	J03	0	125	1553	5684	start	M01
97	T004	J03	0	125	1553	5685	start	M01
97	T005	J03	0	125	1553	5792	start	M03
97	T006	J03	0	125	1553	5685	start	M03
97	T007	J03	0	125	1553	5678	start	M03
97	T008	J03	0	125	1553	5684	start	M03
97	T009	J03	0	125	1553	5705	start	M03
97	T010	J03	0	125	1553	5744	start	M03
97	T011	J03	0	125	1553	5722	start	M04
97	T012	J03	0	125	1553	5664	start	M04
97	T013	J03	0	125	1553	5837	start	M04
97	T014	J03	0	125	1553	5700	start	M04
97	T015	J03	0	125	1553	5681	start	M05
97	T016	J03	0	125	1553	5721	start	M05
97	T017	J03	0	125	1553	5804	start	M05
97	T018	J03	0	125	1553	5705	start	M05
104	T019	J04	0	6250	1244	69	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
104	T020	J05	9	4688	6360	544	submit	-
104	T010	J03	0	125	1553	5744	evict	M03
104	T009	J03	0	125	1553	5705	evict	M03
104	T008	J03	0	125	1553	5684	evict	M03
104	T007	J03	0	125	1553	5678	evict	M03
104	T006	J03	0	125	1553	5685	evict	M03
104	T020	J05	9	4688	6360	544	start	M03
104	T006	J03	0	125	1553	5685	start	M06
104	T007	J03	0	125	1553	5678	start	M06
104	T008	J03	0	125	1553	5684	start	M06
104	T009	J03	0	125	1553	5705	start	M06
104	T010	J03	0	125	1553	5744	start	M07
144	T021	J06	1	2499	311	329	submit	-
144	T021	J06	1	2499	311	329	start	M07
189	T022	J07	6	625	280	36	submit	-
189	T022	J07	6	625	280	36	start	M02
217	T023	J08	4	3125	777	209	submit	-
217	T024	J08	4	3125	777	207	submit	-
217	T025	J08	4	3125	777	206	submit	-
217	T026	J08	4	3125	777	208	submit	-
217	T023	J08	4	3125	777	209	start	M07
217	T024	J08	4	3125	777	207	start	M08
225	T022	J07	6	625	280	36	finish	M02
243	T027	J09	6	625	187	18	submit	-
243	T027	J09	6	625	187	18	start	M02
261	T027	J09	6	625	187	18	finish	M02
284	T028	J10	0	3125	388	272	submit	-
296	T029	J11	6	937	1244	280	submit	-
296	T029	J11	6	937	1244	280	start	M03
298	T030	J12	6	937	221	74	submit	-
298	T031	J12	6	937	221	82	submit	-
298	T032	J12	6	937	221	56	submit	-
298	T033	J12	6	937	221	113	submit	-
298	T030	J12	6	937	221	74	start	M02
343	T034	J13	9	6250	3180	200	submit	-
343	T014	J03	0	125	1553	5700	evict	M04
343	T013	J03	0	125	1553	5837	evict	M04
343	T012	J03	0	125	1553	5664	evict	M04
343	T011	J03	0	125	1553	5722	evict	M04
343	T034	J13	9	6250	3180	200	start	M04
357	T035	J14	4	625	311	31	submit	-
372	T030	J12	6	937	221	74	finish	M02
372	T031	J12	6	937	221	82	start	M02
379	T036	J15	1	1250	466	187	submit	-
419	T037	J16	4	1250	795	465	submit	-
424	T024	J08	4	3125	777	207	finish	M08
424	T032	J12	6	937	221	56	start	M08
424	T033	J12	6	937	221	113	start	M08
424	T035	J14	4	625	311	31	start	M08
424	T011	J03	0	125	1553	5722	start	M08
426	T023	J08	4	3125	777	209	finish	M07
426	T025	J08	4	3125	777	206	start	M07
454	T031	J12	6	937	221	82	finish	M02
454	T036	J15	1	1250	466	187	start	M02
455	T035	J14	4	625	311	31	finish	M08
473	T021	J06	1	2499	311	329	finish	M07
475	T038	J17	1	625	16	735	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
475	T038	J17	1	625	16	735	start	M05
480	T032	J12	6	937	221	56	finish	M08
480	T037	J16	4	1250	795	465	start	M08
518	T015	J03	0	125	1553	5681	fail	M05
518	T016	J03	0	125	1553	5721	fail	M05
518	T017	J03	0	125	1553	5804	fail	M05
518	T018	J03	0	125	1553	5705	fail	M05
518	T038	J17	1	625	16	735	fail	M05
518	T038	J17	1	625	16	735	start	M06
537	T033	J12	6	937	221	113	finish	M08
543	T034	J13	9	6250	3180	200	finish	M04
543	T026	J08	4	3125	777	208	start	M04
543	T012	J03	0	125	1553	5664	start	M04
543	T013	J03	0	125	1553	5837	start	M04
543	T014	J03	0	125	1553	5700	start	M04
566	T039	J18	6	1250	613	317	submit	-
566	T040	J18	6	1250	613	289	submit	-
566	T041	J18	6	1250	613	311	submit	-
566	T042	J18	6	1250	613	312	submit	-
566	T039	J18	6	1250	613	317	start	M04
566	T040	J18	6	1250	613	289	start	M07
566	T041	J18	6	1250	613	311	start	M08
576	T029	J11	6	937	1244	280	finish	M03
576	T042	J18	6	1250	613	312	start	M03
580	T043	J19	6	625	280	57	submit	-
585	T044	J20	0	6250	1244	43	submit	-
632	T025	J08	4	3125	777	206	finish	M07
632	T043	J19	6	625	280	57	start	M07
632	T028	J10	0	3125	388	272	start	M07
641	T036	J15	1	1250	466	187	finish	M02
643	T045	J21	9	6250	233	100	submit	-
643	T004	J03	0	125	1553	5685	evict	M01
643	T003	J03	0	125	1553	5684	evict	M01
643	T045	J21	9	6250	233	100	start	M01
648	T020	J05	9	4688	6360	544	finish	M03
648	T003	J03	0	125	1553	5684	start	M03
648	T004	J03	0	125	1553	5685	start	M03
648	T015	J03	0	125	1553	5681	start	M03
648	T016	J03	0	125	1553	5721	start	M03
650	T046	J22	9	6250	1590	70	submit	-
650	T002	J02	1	4779	2332	5867	evict	M02
650	T046	J22	9	6250	1590	70	start	M02
652	T047	J23	8	625	217	800	submit	-
652	T047	J23	8	625	217	800	start	M03
665	T048	J24	8	625	398	473	submit	-
665	T048	J24	8	625	398	473	start	M03
689	T043	J19	6	625	280	57	finish	M07
691	T049	J25	8	3125	777	744	submit	-
718	T050	J26	6	625	3180	16	submit	-
718	T051	J27	8	5627	1590	208	submit	-
720	T046	J22	9	6250	1590	70	finish	M02
720	T049	J25	8	3125	777	744	start	M02
720	T017	J03	0	125	1553	5804	start	M02
743	T045	J21	9	6250	233	100	finish	M01
743	T051	J27	8	5627	1590	208	start	M01
751	T026	J08	4	3125	777	208	finish	M04
789	T052	J28	6	625	280	43	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
789	T052	J28	6	625	280	43	start	M02
817	T053	J29	4	1250	861	2813	submit	-
817	T054	J29	4	1250	861	2882	submit	-
817	T055	J29	4	1250	861	2315	submit	-
817	T056	J29	4	1250	861	1976	submit	-
817	T053	J29	4	1250	861	2813	start	M04
825	T057	J30	9	4688	6360	531	submit	-
832	T052	J28	6	625	280	43	finish	M02
855	T040	J18	6	1250	613	289	finish	M07
855	T054	J29	4	1250	861	2882	start	M07
877	T041	J18	6	1250	613	311	finish	M08
883	T039	J18	6	1250	613	317	finish	M04
888	T042	J18	6	1250	613	312	finish	M03
888	T016	J03	0	125	1553	5721	evict	M03
888	T015	J03	0	125	1553	5681	evict	M03
888	T004	J03	0	125	1553	5685	evict	M03
888	T003	J03	0	125	1553	5684	evict	M03
888	T057	J30	9	4688	6360	531	start	M03
904	T028	J10	0	3125	388	272	finish	M07
919	T058	J31	6	625	280	360	submit	-
919	T058	J31	6	625	280	360	start	M02
945	T037	J16	4	1250	795	465	finish	M08
945	T055	J29	4	1250	861	2315	start	M08
949	T059	J32	8	5627	1590	172	submit	-
951	T051	J27	8	5627	1590	208	finish	M01
951	T059	J32	8	5627	1590	172	start	M01
976	T060	J33	6	625	280	190	submit	-
976	T060	J33	6	625	280	190	start	M02
1052	T061	J34	4	625	3162	159	submit	-
1052	T062	J34	4	625	3162	618	submit	-
1052	T063	J34	4	625	3162	137	submit	-
1052	T064	J34	4	625	3162	150	submit	-
1063	T065	J35	0	6250	1244	62	submit	-
1123	T059	J32	8	5627	1590	172	finish	M01
1123	T056	J29	4	1250	861	1976	start	M01
1123	T003	J03	0	125	1553	5684	start	M01
1131	T066	J36	6	625	187	13	submit	-
1131	T066	J36	6	625	187	13	start	M01
1138	T048	J24	8	625	398	473	finish	M03
1144	T066	J36	6	625	187	13	finish	M01
1144	T067	J37	8	625	398	1010	submit	-
1144	T067	J37	8	625	398	1010	start	M01
1166	T060	J33	6	625	280	190	finish	M02
1179	T068	J38	6	625	3180	25	submit	-
1183	T069	J39	0	3125	388	263	submit	-
1183	T069	J39	0	3125	388	263	start	M04

Cell window 18: Events

Time 104: task T020 (priority 9) evicted task T006 (priority 0) from machine M03.

Time 104: task T020 (priority 9) evicted task T007 (priority 0) from machine M03.

Time 104: task T020 (priority 9) evicted task T008 (priority 0) from machine M03.

Time 104: task T020 (priority 9) evicted task T009 (priority 0) from machine M03.

Time 104: task T020 (priority 9) evicted task T010 (priority 0) from machine M03.

Time 104: task T006 restarted on machine M06.

Time 104: task T007 restarted on machine M06.

Time 104: task T008 restarted on machine M06.

Time 104: task T009 restarted on machine M06.

Time 104: task T010 restarted on machine M07.

Time 343: task T034 (priority 9) evicted task T011 (priority 0) from machine M04.

Time 343: task T034 (priority 9) evicted task T012 (priority 0) from machine M04.

Time 343: task T034 (priority 9) evicted task T013 (priority 0) from machine M04.

Time 343: task T034 (priority 9) evicted task T014 (priority 0) from machine M04.

Time 424: task T011 restarted on machine M08.

Time 518: machine M05 failed; tasks T015, T016, T017, T018, T038 stopped and went back to the queue.

Time 518: task T038 restarted on machine M06.

Time 543: task T012 restarted on machine M04.

Time 543: task T013 restarted on machine M04.

Time 543: task T014 restarted on machine M04.

Time 643: task T045 (priority 9) evicted task T003 (priority 0) from machine M01.

Time 643: task T045 (priority 9) evicted task T004 (priority 0) from machine M01.

Time 648: task T003 restarted on machine M03.

Time 648: task T004 restarted on machine M03.

Time 648: task T015 restarted on machine M03.

Time 648: task T016 restarted on machine M03.

Time 650: task T046 (priority 9) evicted task T002 (priority 1) from machine M02.

Time 720: task T017 restarted on machine M02.

Time 888: task T057 (priority 9) evicted task T003 (priority 0) from machine M03.

Time 888: task T057 (priority 9) evicted task T004 (priority 0) from machine M03.

Time 888: task T057 (priority 9) evicted task T015 (priority 0) from machine M03.

Time 888: task T057 (priority 9) evicted task T016 (priority 0) from machine M03.

Time 1123: task T003 restarted on machine M01.

Longest waits (submission to first start, tasks started by the window end): task T028 waited 348 s (submitted at 284, first started at 632); task T026 waited 326 s (submitted at 217, first started at 543); task T056 waited 306 s (submitted at 817, first started at 1123); task T025 waited 209 s (submitted at 217, first started at 426); task T055 waited 128 s (submitted at 817, first started at 945).

Cell window 18: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T003	J03	0	125	1553	M01	1123	5607
T056	J29	4	1250	861	M01	1123	1899
T067	J37	8	625	398	M01	1144	954

task	job	priority	CPU	memory	machine	started	remaining
T017	J03	0	125	1553	M02	720	5324
T049	J25	8	3125	777	M02	720	264
T058	J31	6	625	280	M02	919	79
T005	J03	0	125	1553	M03	97	4689
T047	J23	8	625	217	M03	652	252
T057	J30	9	4688	6360	M03	888	219
T012	J03	0	125	1553	M04	543	5007
T013	J03	0	125	1553	M04	543	5180
T014	J03	0	125	1553	M04	543	5043
T053	J29	4	1250	861	M04	817	2430
T069	J39	0	3125	388	M04	1183	246
T006	J03	0	125	1553	M06	104	4589
T007	J03	0	125	1553	M06	104	4582
T008	J03	0	125	1553	M06	104	4588
T009	J03	0	125	1553	M06	104	4609
T038	J17	1	625	16	M06	518	53
T010	J03	0	125	1553	M07	104	4648
T054	J29	4	1250	861	M07	855	2537
T011	J03	0	125	1553	M08	424	4946
T055	J29	4	1250	861	M08	945	2060

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T050	J26	6	625	3180	718	16
T068	J38	6	625	3180	1179	25
T061	J34	4	625	3162	1052	159
T062	J34	4	625	3162	1052	618
T063	J34	4	625	3162	1052	137
T064	J34	4	625	3162	1052	150
T002	J02	1	4779	2332	71	5867
T004	J03	0	125	1553	97	5685
T015	J03	0	125	1553	97	5681
T016	J03	0	125	1553	97	5721
T018	J03	0	125	1553	97	5705
T019	J04	0	6250	1244	104	69
T044	J20	0	6250	1244	585	43
T065	J35	0	6250	1244	1063	62

Machine M05 is down at time 1200; it does not return.

Cell window 22

Times 0 to 1200. The cluster has 8 machines. 127 tasks of 54 jobs are submitted during the window.

Cell window 22: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	3116	-	-
M03	6250	6244	-	-
M04	6250	6244	-	-
M05	6250	3116	493	1435
M06	6250	6244	-	-
M07	6250	6244	-	-
M08	6250	6244	-	-

Cell window 22: Task log

time	task	job	priority	CPU	memory	run time	event	machine
24	T001	J01	9	6250	3180	236	submit	-
24	T001	J01	9	6250	3180	236	start	M03
26	T002	J02	0	6250	466	44	submit	-
26	T002	J02	0	6250	466	44	start	M01
58	T003	J03	1	937	1590	1173	submit	-
58	T003	J03	1	937	1590	1173	start	M02
60	T004	J04	0	1250	919	1171	submit	-
60	T004	J04	0	1250	919	1171	start	M02
62	T005	J05	0	937	244	81	submit	-
62	T006	J05	0	937	244	72	submit	-
62	T007	J05	0	937	244	77	submit	-
62	T008	J05	0	937	244	77	submit	-
62	T009	J05	0	937	244	82	submit	-
62	T010	J05	0	937	244	72	submit	-
62	T011	J05	0	937	244	90	submit	-
62	T012	J05	0	937	244	72	submit	-
62	T013	J05	0	937	244	77	submit	-
62	T014	J05	0	937	244	81	submit	-
62	T015	J05	0	937	244	77	submit	-
62	T016	J05	0	937	244	78	submit	-
62	T017	J05	0	937	244	70	submit	-
62	T018	J05	0	937	244	108	submit	-
62	T019	J05	0	937	244	89	submit	-
62	T020	J05	0	937	244	73	submit	-
62	T005	J05	0	937	244	81	start	M02
62	T006	J05	0	937	244	72	start	M02
62	T007	J05	0	937	244	77	start	M04
62	T008	J05	0	937	244	77	start	M04
62	T009	J05	0	937	244	82	start	M04
62	T010	J05	0	937	244	72	start	M04
62	T011	J05	0	937	244	90	start	M04
62	T012	J05	0	937	244	72	start	M04
62	T013	J05	0	937	244	77	start	M05
62	T014	J05	0	937	244	81	start	M05
62	T015	J05	0	937	244	77	start	M05
62	T016	J05	0	937	244	78	start	M05
62	T017	J05	0	937	244	70	start	M05
62	T018	J05	0	937	244	108	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
62	T019	J05	0	937	244	89	start	M06
62	T020	J05	0	937	244	73	start	M06
70	T002	J02	0	6250	466	44	finish	M01
132	T017	J05	0	937	244	70	finish	M05
134	T006	J05	0	937	244	72	finish	M02
134	T010	J05	0	937	244	72	finish	M04
134	T012	J05	0	937	244	72	finish	M04
135	T020	J05	0	937	244	73	finish	M06
139	T007	J05	0	937	244	77	finish	M04
139	T008	J05	0	937	244	77	finish	M04
139	T013	J05	0	937	244	77	finish	M05
139	T015	J05	0	937	244	77	finish	M05
140	T016	J05	0	937	244	78	finish	M05
141	T021	J06	0	6250	1244	58	submit	-
141	T021	J06	0	6250	1244	58	start	M01
142	T022	J07	0	6250	1244	58	submit	-
142	T022	J07	0	6250	1244	58	start	M07
143	T005	J05	0	937	244	81	finish	M02
143	T014	J05	0	937	244	81	finish	M05
144	T009	J05	0	937	244	82	finish	M04
145	T023	J08	9	6250	3528	2929	submit	-
145	T023	J08	9	6250	3528	2929	start	M08
151	T019	J05	0	937	244	89	finish	M06
152	T011	J05	0	937	244	90	finish	M04
170	T018	J05	0	937	244	108	finish	M05
195	T024	J09	4	2328	2704	756	submit	-
195	T025	J09	4	2328	2704	785	submit	-
195	T026	J09	4	2328	2704	749	submit	-
195	T027	J09	4	2328	2704	817	submit	-
195	T028	J10	4	2975	2704	695	submit	-
195	T029	J10	4	2975	2704	698	submit	-
195	T030	J10	4	2975	2704	698	submit	-
195	T031	J10	4	2975	2704	697	submit	-
195	T024	J09	4	2328	2704	756	start	M04
195	T025	J09	4	2328	2704	785	start	M04
195	T026	J09	4	2328	2704	749	start	M05
195	T027	J09	4	2328	2704	817	start	M06
195	T028	J10	4	2975	2704	695	start	M06
199	T021	J06	0	6250	1244	58	finish	M01
199	T029	J10	4	2975	2704	698	start	M01
200	T022	J07	0	6250	1244	58	finish	M07
200	T030	J10	4	2975	2704	698	start	M07
200	T031	J10	4	2975	2704	697	start	M07
224	T032	J11	4	1250	155	12	submit	-
224	T032	J11	4	1250	155	12	start	M01
236	T032	J11	4	1250	155	12	finish	M01
260	T001	J01	9	6250	3180	236	finish	M03
273	T033	J12	4	3125	1125	75	submit	-
273	T033	J12	4	3125	1125	75	start	M03
315	T034	J13	4	1874	1540	456	submit	-
315	T035	J13	4	1874	1540	446	submit	-
315	T036	J13	4	1874	1540	449	submit	-
315	T037	J13	4	1874	1540	465	submit	-
315	T034	J13	4	1874	1540	456	start	M03
323	T038	J14	9	6250	233	114	submit	-
323	T004	J04	0	1250	919	1171	evict	M02
323	T003	J03	1	937	1590	1173	evict	M02

time	task	job	priority	CPU	memory	run time	event	machine
323	T038	J14	9	6250	233	114	start	M02
323	T003	J03	1	937	1590	1173	start	M03
339	T039	J15	4	3125	1125	42	submit	-
348	T033	J12	4	3125	1125	75	finish	M03
348	T035	J13	4	1874	1540	446	start	M03
348	T004	J04	0	1250	919	1171	start	M03
393	T040	J16	4	3125	1125	80	submit	-
437	T038	J14	9	6250	233	114	finish	M02
437	T036	J13	4	1874	1540	449	start	M02
437	T037	J13	4	1874	1540	465	start	M02
454	T041	J17	8	5627	1590	268	submit	-
493	T026	J09	4	2328	2704	749	fail	M05
521	T042	J18	8	625	16	28	submit	-
521	T042	J18	8	625	16	28	start	M01
549	T042	J18	8	625	16	28	finish	M01
573	T043	J19	8	3125	388	33	submit	-
573	T043	J19	8	3125	388	33	start	M01
606	T043	J19	8	3125	388	33	finish	M01
621	T044	J20	0	6250	466	44	submit	-
622	T045	J21	0	6250	1244	65	submit	-
622	T046	J22	0	6250	1244	53	submit	-
623	T047	J23	1	1250	1590	365	submit	-
630	T048	J24	4	625	155	34	submit	-
630	T049	J25	4	625	155	28	submit	-
630	T048	J24	4	625	155	34	start	M01
630	T049	J25	4	625	155	28	start	M01
653	T050	J26	1	1250	466	200	submit	-
653	T050	J26	1	1250	466	200	start	M04
655	T051	J27	0	625	227	198	submit	-
655	T052	J27	0	625	227	198	submit	-
655	T053	J27	0	625	227	197	submit	-
655	T054	J27	0	625	227	198	submit	-
655	T055	J27	0	625	227	198	submit	-
655	T056	J27	0	625	227	197	submit	-
655	T057	J27	0	625	227	197	submit	-
655	T058	J27	0	625	227	197	submit	-
655	T059	J27	0	625	227	198	submit	-
655	T060	J27	0	625	227	197	submit	-
655	T061	J27	0	625	227	197	submit	-
655	T062	J27	0	625	227	197	submit	-
655	T063	J27	0	625	227	198	submit	-
655	T064	J27	0	625	227	197	submit	-
655	T051	J27	0	625	227	198	start	M06
657	T065	J28	4	1250	466	365	submit	-
658	T049	J25	4	625	155	28	finish	M01
658	T052	J27	0	625	227	198	start	M01
661	T066	J29	0	937	1193	896	submit	-
661	T067	J29	0	937	1193	897	submit	-
661	T068	J29	0	937	1193	894	submit	-
661	T069	J29	0	937	1193	895	submit	-
661	T070	J29	0	937	1193	891	submit	-
661	T071	J29	0	937	1193	895	submit	-
661	T072	J29	0	937	1193	897	submit	-
661	T073	J29	0	937	1193	899	submit	-
661	T074	J29	0	937	1193	897	submit	-
661	T075	J29	0	937	1193	896	submit	-
661	T076	J29	0	937	1193	895	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
661	T077	J29	0	937	1193	897	submit	-
661	T078	J29	0	937	1193	895	submit	-
661	T079	J29	0	937	1193	896	submit	-
661	T080	J29	0	937	1193	894	submit	-
661	T081	J29	0	937	1193	890	submit	-
664	T048	J24	4	625	155	34	finish	M01
771	T034	J13	4	1874	1540	456	finish	M03
771	T065	J28	4	1250	466	365	start	M03
771	T053	J27	0	625	227	197	start	M03
794	T035	J13	4	1874	1540	446	finish	M03
794	T047	J23	1	1250	1590	365	start	M03
794	T054	J27	0	625	227	198	start	M03
806	T082	J30	1	1562	1553	488	submit	-
814	T083	J31	1	1562	1553	95	submit	-
820	T084	J32	1	1562	1553	96	submit	-
845	T085	J33	1	1250	1553	53	submit	-
846	T086	J34	0	1250	197	52	submit	-
848	T087	J35	1	1250	1553	433	submit	-
851	T088	J36	0	1250	197	273	submit	-
851	T089	J36	0	1250	197	429	submit	-
853	T050	J26	1	1250	466	200	finish	M04
853	T051	J27	0	625	227	198	finish	M06
853	T090	J37	1	1250	1553	52	submit	-
853	T055	J27	0	625	227	198	start	M04
853	T056	J27	0	625	227	197	start	M04
853	T057	J27	0	625	227	197	start	M06
854	T091	J38	0	1250	197	51	submit	-
856	T052	J27	0	625	227	198	finish	M01
861	T092	J39	0	3125	388	286	submit	-
886	T036	J13	4	1874	1540	449	finish	M02
886	T039	J15	4	3125	1125	42	start	M02
886	T058	J27	0	625	227	197	start	M02
890	T028	J10	4	2975	2704	695	finish	M06
890	T026	J09	4	2328	2704	749	start	M06
890	T059	J27	0	625	227	198	start	M06
897	T029	J10	4	2975	2704	698	finish	M01
897	T031	J10	4	2975	2704	697	finish	M07
897	T041	J17	8	5627	1590	268	start	M01
897	T040	J16	4	3125	1125	80	start	M07
898	T030	J10	4	2975	2704	698	finish	M07
898	T082	J30	1	1562	1553	488	start	M07
898	T083	J31	1	1562	1553	95	start	M07
902	T037	J13	4	1874	1540	465	finish	M02
902	T084	J32	1	1562	1553	96	start	M02
921	T093	J40	0	1250	16	21	submit	-
924	T094	J41	9	6250	3180	205	submit	-
928	T039	J15	4	3125	1125	42	finish	M02
928	T060	J27	0	625	227	197	start	M02
928	T061	J27	0	625	227	197	start	M02
928	T062	J27	0	625	227	197	start	M02
928	T063	J27	0	625	227	198	start	M02
928	T064	J27	0	625	227	197	start	M02
951	T024	J09	4	2328	2704	756	finish	M04
951	T085	J33	1	1250	1553	53	start	M04
951	T086	J34	0	1250	197	52	start	M04
968	T053	J27	0	625	227	197	finish	M03
968	T066	J29	0	937	1193	896	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
972	T095	J42	4	1250	155	21	submit	-
972	T096	J43	4	1250	466	299	submit	-
972	T097	J44	8	3125	1590	563	submit	-
973	T098	J45	4	1874	1590	534	submit	-
973	T099	J45	4	1874	1590	537	submit	-
973	T100	J45	4	1874	1590	535	submit	-
973	T101	J45	4	1874	1590	558	submit	-
977	T040	J16	4	3125	1125	80	finish	M07
977	T083	J31	1	1562	1553	95	evict	M07
977	T082	J30	1	1562	1553	488	evict	M07
977	T094	J41	9	6250	3180	205	start	M07
980	T025	J09	4	2328	2704	785	finish	M04
980	T095	J42	4	1250	155	21	start	M04
980	T093	J40	0	1250	16	21	start	M04
992	T054	J27	0	625	227	198	finish	M03
998	T084	J32	1	1562	1553	96	finish	M02
998	T096	J43	4	1250	466	299	start	M02
998	T067	J29	0	937	1193	897	start	M02
1001	T093	J40	0	1250	16	21	finish	M04
1001	T095	J42	4	1250	155	21	finish	M04
1001	T102	J46	4	1250	155	355	submit	-
1003	T086	J34	0	1250	197	52	finish	M04
1003	T097	J44	8	3125	1590	563	start	M04
1004	T085	J33	1	1250	1553	53	finish	M04
1004	T098	J45	4	1874	1590	534	start	M04
1012	T027	J09	4	2328	2704	817	finish	M06
1012	T099	J45	4	1874	1590	537	start	M06
1050	T056	J27	0	625	227	197	finish	M04
1050	T057	J27	0	625	227	197	finish	M06
1051	T055	J27	0	625	227	198	finish	M04
1051	T102	J46	4	1250	155	355	start	M04
1083	T058	J27	0	625	227	197	finish	M02
1088	T059	J27	0	625	227	198	finish	M06
1088	T100	J45	4	1874	1590	535	start	M06
1101	T103	J47	0	6250	1244	55	submit	-
1102	T104	J48	0	6250	1244	46	submit	-
1107	T105	J49	9	62	466	1897	submit	-
1107	T105	J49	9	62	466	1897	start	M01
1112	T106	J50	4	3125	1125	35	submit	-
1125	T060	J27	0	625	227	197	finish	M02
1125	T061	J27	0	625	227	197	finish	M02
1125	T062	J27	0	625	227	197	finish	M02
1125	T064	J27	0	625	227	197	finish	M02
1125	T106	J50	4	3125	1125	35	start	M02
1126	T063	J27	0	625	227	198	finish	M02
1136	T065	J28	4	1250	466	365	finish	M03
1136	T088	J36	0	1250	197	273	start	M03
1143	T107	J51	9	909	514	1842	submit	-
1143	T067	J29	0	937	1193	897	evict	M02
1143	T107	J51	9	909	514	1842	start	M02
1159	T047	J23	1	1250	1590	365	finish	M03
1159	T101	J45	4	1874	1590	558	start	M03
1160	T106	J50	4	3125	1125	35	finish	M02
1160	T082	J30	1	1562	1553	488	start	M02
1160	T089	J36	0	1250	197	429	start	M02
1160	T091	J38	0	1250	197	51	start	M02
1165	T041	J17	8	5627	1590	268	finish	M01

time	task	job	priority	CPU	memory	run time	event	machine
1165	T083	J31	1	1562	1553	95	start	M01
1165	T092	J39	0	3125	388	286	start	M01
1166	T108	J52	1	1353	777	1712	submit	-
1169	T109	J53	0	1250	777	1140	submit	-
1169	T110	J53	0	1250	777	1709	submit	-
1169	T111	J53	0	1250	777	1202	submit	-
1170	T112	J54	0	937	267	100	submit	-
1170	T113	J54	0	937	267	149	submit	-
1170	T114	J54	0	937	267	106	submit	-
1170	T115	J54	0	937	267	109	submit	-
1170	T116	J54	0	937	267	110	submit	-
1170	T117	J54	0	937	267	102	submit	-
1170	T118	J54	0	937	267	108	submit	-
1170	T119	J54	0	937	267	102	submit	-
1170	T120	J54	0	937	267	100	submit	-
1170	T121	J54	0	937	267	109	submit	-
1170	T122	J54	0	937	267	101	submit	-
1170	T123	J54	0	937	267	100	submit	-
1170	T124	J54	0	937	267	89	submit	-
1170	T125	J54	0	937	267	101	submit	-
1170	T126	J54	0	937	267	89	submit	-
1170	T127	J54	0	937	267	93	submit	-
1170	T112	J54	0	937	267	100	start	M01
1182	T094	J41	9	6250	3180	205	finish	M07
1182	T087	J35	1	1250	1553	433	start	M07
1182	T090	J37	1	1250	1553	52	start	M07
1182	T108	J52	1	1353	777	1712	start	M07
1182	T067	J29	0	937	1193	897	start	M07
1182	T109	J53	0	1250	777	1140	start	M07

Cell window 22: Events

Time 323: task T038 (priority 9) evicted task T003 (priority 1) from machine M02.

Time 323: task T038 (priority 9) evicted task T004 (priority 0) from machine M02.

Time 323: task T003 restarted on machine M03.

Time 348: task T004 restarted on machine M03.

Time 493: machine M05 failed; task T026 stopped and went back to the queue.

Time 890: task T026 restarted on machine M06.

Time 977: task T094 (priority 9) evicted task T082 (priority 1) from machine M07.

Time 977: task T094 (priority 9) evicted task T083 (priority 1) from machine M07.

Time 1143: task T107 (priority 9) evicted task T067 (priority 0) from machine M02.

Time 1160: task T082 restarted on machine M02.

Time 1165: task T083 restarted on machine M01.

Time 1182: task T067 restarted on machine M07.

Longest waits (submission to first start, tasks started by the window end): task T039 waited 547 s (submitted at 339, first started at 886); task T040 waited 504 s (submitted at 393, first started at 897); task T041 waited 443 s (submitted at 454, first started at 897); task T067 waited 337 s (submitted at 661, first started at 998); task T087 waited 334 s (submitted at 848, first started at 1182).

Cell window 22: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T083	J31	1	1562	1553	M01	1165	60
T092	J39	0	3125	388	M01	1165	251
T105	J49	9	62	466	M01	1107	1804
T112	J54	0	937	267	M01	1170	70
T082	J30	1	1562	1553	M02	1160	448
T089	J36	0	1250	197	M02	1160	389
T091	J38	0	1250	197	M02	1160	11
T096	J43	4	1250	466	M02	998	97
T107	J51	9	909	514	M02	1143	1785
T003	J03	1	937	1590	M03	323	296
T004	J04	0	1250	919	M03	348	319
T066	J29	0	937	1193	M03	968	664
T088	J36	0	1250	197	M03	1136	209
T101	J45	4	1874	1590	M03	1159	517
T097	J44	8	3125	1590	M04	1003	366
T098	J45	4	1874	1590	M04	1004	338
T102	J46	4	1250	155	M04	1051	206
T026	J09	4	2328	2704	M06	890	439
T099	J45	4	1874	1590	M06	1012	349
T100	J45	4	1874	1590	M06	1088	423
T067	J29	0	937	1193	M07	1182	879
T087	J35	1	1250	1553	M07	1182	415
T090	J37	1	1250	1553	M07	1182	34
T108	J52	1	1353	777	M07	1182	1694
T109	J53	0	1250	777	M07	1182	1122
T023	J08	9	6250	3528	M08	145	1874

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T044	J20	0	6250	466	621	44
T045	J21	0	6250	1244	622	65
T046	J22	0	6250	1244	622	53
T068	J29	0	937	1193	661	894
T069	J29	0	937	1193	661	895
T070	J29	0	937	1193	661	891
T071	J29	0	937	1193	661	895
T072	J29	0	937	1193	661	897
T073	J29	0	937	1193	661	899
T074	J29	0	937	1193	661	897
T075	J29	0	937	1193	661	896
T076	J29	0	937	1193	661	895
T077	J29	0	937	1193	661	897
T078	J29	0	937	1193	661	895
T079	J29	0	937	1193	661	896
T080	J29	0	937	1193	661	894
T081	J29	0	937	1193	661	890
T103	J47	0	6250	1244	1101	55
T104	J48	0	6250	1244	1102	46
T110	J53	0	1250	777	1169	1709
T111	J53	0	1250	777	1169	1202
T113	J54	0	937	267	1170	149
T114	J54	0	937	267	1170	106
T115	J54	0	937	267	1170	109
T116	J54	0	937	267	1170	110

task	job	priority	CPU	memory	submitted	run time
T117	J54	0	937	267	1170	102
T118	J54	0	937	267	1170	108
T119	J54	0	937	267	1170	102
T120	J54	0	937	267	1170	100
T121	J54	0	937	267	1170	109
T122	J54	0	937	267	1170	101
T123	J54	0	937	267	1170	100
T124	J54	0	937	267	1170	89
T125	J54	0	937	267	1170	101
T126	J54	0	937	267	1170	89
T127	J54	0	937	267	1170	93

Machine M05 is down at time 1200; it returns at time 1435.

Cell window 25

Times 0 to 1200. The cluster has 8 machines. 160 tasks of 62 jobs are submitted during the window.

Cell window 25: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	579	934
M03	6250	6244	-	-
M04	6250	6244	-	-
M05	12500	12500	-	-
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	6250	6244	-	-

Cell window 25: Task log

time	task	job	priority	CPU	memory	run time	event	machine
1	T001	J01	1	1250	777	1057	submit	-
1	T001	J01	1	1250	777	1057	start	M01
20	T002	J02	0	3125	466	310	submit	-
20	T002	J02	0	3125	466	310	start	M01
51	T003	J03	0	1688	295	996	submit	-
51	T003	J03	0	1688	295	996	start	M01
59	T004	J04	6	625	280	157	submit	-
59	T004	J04	6	625	280	157	start	M02
80	T005	J05	0	1250	16	18	submit	-
80	T005	J05	0	1250	16	18	start	M02
88	T006	J06	4	3125	448	20	submit	-
88	T006	J06	4	3125	448	20	start	M02
98	T005	J05	0	1250	16	18	finish	M02
108	T006	J06	4	3125	448	20	finish	M02
141	T007	J07	8	1874	777	3413	submit	-
141	T007	J07	8	1874	777	3413	start	M02
200	T008	J08	4	3381	1413	459	submit	-
200	T009	J08	4	3381	1413	485	submit	-
200	T010	J08	4	3381	1413	481	submit	-
200	T011	J08	4	3381	1413	499	submit	-
200	T008	J08	4	3381	1413	459	start	M02
200	T009	J08	4	3381	1413	485	start	M03
200	T010	J08	4	3381	1413	481	start	M04
200	T011	J08	4	3381	1413	499	start	M05
216	T004	J04	6	625	280	157	finish	M02
223	T012	J09	1	1250	777	1094	submit	-
223	T012	J09	1	1250	777	1094	start	M03
227	T013	J10	6	625	280	229	submit	-
227	T013	J10	6	625	280	229	start	M02
260	T014	J11	0	6250	1244	62	submit	-
260	T014	J11	0	6250	1244	62	start	M05
261	T015	J12	0	6250	1244	83	submit	-
261	T015	J12	0	6250	1244	83	start	M06
271	T016	J13	0	1688	295	1034	submit	-
271	T016	J13	0	1688	295	1034	start	M04
272	T017	J14	6	937	253	50	submit	-
272	T018	J14	6	937	253	45	submit	-
272	T019	J14	6	937	253	44	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
272	T020	J14	6	937	253	169	submit	-
272	T017	J14	6	937	253	50	start	M03
272	T018	J14	6	937	253	45	start	M04
272	T019	J14	6	937	253	44	start	M05
272	T020	J14	6	937	253	169	start	M05
290	T021	J15	1	937	777	75	submit	-
290	T021	J15	1	937	777	75	start	M05
292	T022	J16	0	1250	777	73	submit	-
292	T023	J16	0	1250	777	73	submit	-
292	T024	J16	0	1250	777	71	submit	-
292	T022	J16	0	1250	777	73	start	M07
292	T023	J16	0	1250	777	73	start	M07
292	T024	J16	0	1250	777	71	start	M07
293	T025	J17	0	937	277	54	submit	-
293	T026	J17	0	937	277	53	submit	-
293	T027	J17	0	937	277	54	submit	-
293	T028	J17	0	937	277	53	submit	-
293	T029	J17	0	937	277	64	submit	-
293	T030	J17	0	937	277	63	submit	-
293	T031	J17	0	937	277	67	submit	-
293	T032	J17	0	937	277	53	submit	-
293	T033	J17	0	937	277	63	submit	-
293	T034	J17	0	937	277	51	submit	-
293	T035	J17	0	937	277	54	submit	-
293	T036	J17	0	937	277	57	submit	-
293	T037	J17	0	937	277	55	submit	-
293	T038	J17	0	937	277	55	submit	-
293	T039	J17	0	937	277	51	submit	-
293	T040	J17	0	937	277	51	submit	-
293	T025	J17	0	937	277	54	start	M07
293	T026	J17	0	937	277	53	start	M07
293	T027	J17	0	937	277	54	start	M08
293	T028	J17	0	937	277	53	start	M08
293	T029	J17	0	937	277	64	start	M08
293	T030	J17	0	937	277	63	start	M08
293	T031	J17	0	937	277	67	start	M08
293	T032	J17	0	937	277	53	start	M08
314	T041	J18	6	625	187	22	submit	-
314	T041	J18	6	625	187	22	start	M03
316	T019	J14	6	937	253	44	finish	M05
316	T033	J17	0	937	277	63	start	M05
317	T018	J14	6	937	253	45	finish	M04
317	T034	J17	0	937	277	51	start	M04
322	T014	J11	0	6250	1244	62	finish	M05
322	T017	J14	6	937	253	50	finish	M03
322	T035	J17	0	937	277	54	start	M03
322	T036	J17	0	937	277	57	start	M05
322	T037	J17	0	937	277	55	start	M05
322	T038	J17	0	937	277	55	start	M05
322	T039	J17	0	937	277	51	start	M05
322	T040	J17	0	937	277	51	start	M05
330	T002	J02	0	3125	466	310	finish	M01
336	T041	J18	6	625	187	22	finish	M03
344	T015	J12	0	6250	1244	83	finish	M06
346	T026	J17	0	937	277	53	finish	M07
346	T028	J17	0	937	277	53	finish	M08
346	T032	J17	0	937	277	53	finish	M08

time	task	job	priority	CPU	memory	run time	event	machine
347	T025	J17	0	937	277	54	finish	M07
347	T027	J17	0	937	277	54	finish	M08
356	T030	J17	0	937	277	63	finish	M08
357	T029	J17	0	937	277	64	finish	M08
360	T031	J17	0	937	277	67	finish	M08
363	T024	J16	0	1250	777	71	finish	M07
365	T021	J15	1	937	777	75	finish	M05
365	T022	J16	0	1250	777	73	finish	M07
365	T023	J16	0	1250	777	73	finish	M07
368	T034	J17	0	937	277	51	finish	M04
373	T039	J17	0	937	277	51	finish	M05
373	T040	J17	0	937	277	51	finish	M05
376	T035	J17	0	937	277	54	finish	M03
377	T037	J17	0	937	277	55	finish	M05
377	T038	J17	0	937	277	55	finish	M05
379	T033	J17	0	937	277	63	finish	M05
379	T036	J17	0	937	277	57	finish	M05
380	T042	J19	9	6250	233	115	submit	-
380	T042	J19	9	6250	233	115	start	M05
382	T043	J20	0	6250	466	34	submit	-
382	T043	J20	0	6250	466	34	start	M06
391	T044	J21	6	937	1244	71	submit	-
391	T045	J22	8	625	466	1395	submit	-
391	T046	J23	4	625	311	13	submit	-
391	T045	J22	8	625	466	1395	start	M01
391	T044	J21	6	937	1244	71	start	M01
391	T046	J23	4	625	311	13	start	M01
395	T047	J24	9	6250	1590	80	submit	-
395	T047	J24	9	6250	1590	80	start	M07
404	T046	J23	4	625	311	13	finish	M01
416	T043	J20	0	6250	466	34	finish	M06
441	T020	J14	6	937	253	169	finish	M05
456	T013	J10	6	625	280	229	finish	M02
462	T044	J21	6	937	1244	71	finish	M01
474	T048	J25	6	937	1244	185	submit	-
474	T048	J25	6	937	1244	185	start	M01
475	T047	J24	9	6250	1590	80	finish	M07
495	T042	J19	9	6250	233	115	finish	M05
500	T049	J26	9	4688	6360	491	submit	-
500	T049	J26	9	4688	6360	491	start	M05
504	T050	J27	6	937	1244	68	submit	-
504	T050	J27	6	937	1244	68	start	M02
505	T051	J28	6	937	197	66	submit	-
505	T052	J28	6	937	197	68	submit	-
505	T053	J28	6	937	197	64	submit	-
505	T054	J28	6	937	197	63	submit	-
505	T051	J28	6	937	197	66	start	M01
505	T052	J28	6	937	197	68	start	M03
505	T053	J28	6	937	197	64	start	M04
505	T054	J28	6	937	197	63	start	M05
567	T055	J29	6	6250	622	115	submit	-
567	T056	J30	6	625	280	55	submit	-
567	T055	J29	6	6250	622	115	start	M06
567	T056	J30	6	625	280	55	start	M03
568	T054	J28	6	937	197	63	finish	M05
569	T053	J28	6	937	197	64	finish	M04
571	T051	J28	6	937	197	66	finish	M01

time	task	job	priority	CPU	memory	run time	event	machine
572	T050	J27	6	937	1244	68	finish	M02
573	T052	J28	6	937	197	68	finish	M03
573	T057	J31	1	937	622	236	submit	-
573	T057	J31	1	937	622	236	start	M02
575	T058	J32	0	1250	622	130	submit	-
575	T059	J32	0	1250	622	116	submit	-
575	T060	J32	0	1250	622	115	submit	-
575	T061	J32	0	1250	622	141	submit	-
575	T062	J32	0	1250	622	195	submit	-
575	T063	J32	0	1250	622	139	submit	-
575	T064	J32	0	1250	622	162	submit	-
575	T065	J32	0	1250	622	176	submit	-
575	T066	J32	0	1250	622	117	submit	-
575	T067	J32	0	1250	622	119	submit	-
575	T068	J32	0	1250	622	157	submit	-
575	T069	J32	0	1250	622	119	submit	-
575	T070	J32	0	1250	622	170	submit	-
575	T071	J32	0	1250	622	119	submit	-
575	T072	J32	0	1250	622	180	submit	-
575	T073	J32	0	1250	622	126	submit	-
575	T058	J32	0	1250	622	130	start	M05
575	T059	J32	0	1250	622	116	start	M05
575	T060	J32	0	1250	622	115	start	M05
575	T061	J32	0	1250	622	141	start	M07
575	T062	J32	0	1250	622	195	start	M07
575	T063	J32	0	1250	622	139	start	M07
575	T064	J32	0	1250	622	162	start	M07
575	T065	J32	0	1250	622	176	start	M07
575	T066	J32	0	1250	622	117	start	M08
575	T067	J32	0	1250	622	119	start	M08
575	T068	J32	0	1250	622	157	start	M08
575	T069	J32	0	1250	622	119	start	M08
575	T070	J32	0	1250	622	170	start	M08
579	T007	J07	8	1874	777	3413	fail	M02
579	T008	J08	4	3381	1413	459	fail	M02
579	T057	J31	1	937	622	236	fail	M02
579	T057	J31	1	937	622	236	start	M03
622	T056	J30	6	625	280	55	finish	M03
659	T048	J25	6	937	1244	185	finish	M01
659	T007	J07	8	1874	777	3413	start	M01
681	T010	J08	4	3381	1413	481	finish	M04
681	T008	J08	4	3381	1413	459	start	M04
682	T055	J29	6	6250	622	115	finish	M06
682	T074	J33	6	625	280	129	submit	-
682	T074	J33	6	625	280	129	start	M01
682	T071	J32	0	1250	622	119	start	M06
682	T072	J32	0	1250	622	180	start	M06
682	T073	J32	0	1250	622	126	start	M06
685	T009	J08	4	3381	1413	485	finish	M03
690	T060	J32	0	1250	622	115	finish	M05
691	T059	J32	0	1250	622	116	finish	M05
692	T066	J32	0	1250	622	117	finish	M08
694	T067	J32	0	1250	622	119	finish	M08
694	T069	J32	0	1250	622	119	finish	M08
699	T011	J08	4	3381	1413	499	finish	M05
705	T058	J32	0	1250	622	130	finish	M05
714	T063	J32	0	1250	622	139	finish	M07

time	task	job	priority	CPU	memory	run time	event	machine
716	T061	J32	0	1250	622	141	finish	M07
720	T075	J34	4	625	155	36	submit	-
720	T075	J34	4	625	155	36	start	M03
721	T076	J35	6	645	197	74	submit	-
721	T077	J35	6	645	197	76	submit	-
721	T076	J35	6	645	197	74	start	M03
721	T077	J35	6	645	197	76	start	M03
732	T068	J32	0	1250	622	157	finish	M08
737	T064	J32	0	1250	622	162	finish	M07
742	T078	J36	0	6250	1244	53	submit	-
742	T078	J36	0	6250	1244	53	start	M05
745	T070	J32	0	1250	622	170	finish	M08
745	T079	J37	0	6250	1244	51	submit	-
745	T079	J37	0	6250	1244	51	start	M08
751	T065	J32	0	1250	622	176	finish	M07
756	T075	J34	4	625	155	36	finish	M03
769	T080	J38	6	937	258	199	submit	-
769	T081	J38	6	937	258	113	submit	-
769	T082	J38	6	937	258	46	submit	-
769	T083	J38	6	937	258	42	submit	-
769	T080	J38	6	937	258	199	start	M03
769	T081	J38	6	937	258	113	start	M03
769	T082	J38	6	937	258	46	start	M04
769	T083	J38	6	937	258	42	start	M05
770	T062	J32	0	1250	622	195	finish	M07
794	T084	J39	1	937	777	86	submit	-
794	T084	J39	1	937	777	86	start	M06
795	T076	J35	6	645	197	74	finish	M03
795	T078	J36	0	6250	1244	53	finish	M05
796	T079	J37	0	6250	1244	51	finish	M08
796	T085	J40	0	1250	777	83	submit	-
796	T086	J40	0	1250	777	80	submit	-
796	T087	J40	0	1250	777	82	submit	-
796	T088	J40	0	1250	777	84	submit	-
796	T089	J40	0	1250	777	83	submit	-
796	T090	J40	0	1250	777	74	submit	-
796	T091	J40	0	1250	777	73	submit	-
796	T092	J40	0	1250	777	82	submit	-
796	T093	J40	0	1250	777	81	submit	-
796	T094	J40	0	1250	777	71	submit	-
796	T085	J40	0	1250	777	83	start	M03
796	T086	J40	0	1250	777	80	start	M05
796	T087	J40	0	1250	777	82	start	M05
796	T088	J40	0	1250	777	84	start	M05
796	T089	J40	0	1250	777	83	start	M05
796	T090	J40	0	1250	777	74	start	M05
796	T091	J40	0	1250	777	73	start	M06
796	T092	J40	0	1250	777	82	start	M07
796	T093	J40	0	1250	777	81	start	M07
796	T094	J40	0	1250	777	71	start	M07
797	T077	J35	6	645	197	76	finish	M03
798	T095	J41	0	937	277	58	submit	-
798	T096	J41	0	937	277	49	submit	-
798	T097	J41	0	937	277	50	submit	-
798	T098	J41	0	937	277	48	submit	-
798	T099	J41	0	937	277	47	submit	-
798	T100	J41	0	937	277	49	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
798	T101	J41	0	937	277	74	submit	-
798	T102	J41	0	937	277	58	submit	-
798	T103	J41	0	937	277	52	submit	-
798	T104	J41	0	937	277	47	submit	-
798	T105	J41	0	937	277	51	submit	-
798	T106	J41	0	937	277	63	submit	-
798	T107	J41	0	937	277	47	submit	-
798	T108	J41	0	937	277	46	submit	-
798	T109	J41	0	937	277	48	submit	-
798	T110	J41	0	937	277	47	submit	-
798	T095	J41	0	937	277	58	start	M03
798	T096	J41	0	937	277	49	start	M07
798	T097	J41	0	937	277	50	start	M07
798	T098	J41	0	937	277	48	start	M08
798	T099	J41	0	937	277	47	start	M08
798	T100	J41	0	937	277	49	start	M08
798	T101	J41	0	937	277	74	start	M08
798	T102	J41	0	937	277	58	start	M08
798	T103	J41	0	937	277	52	start	M08
801	T071	J32	0	1250	622	119	finish	M06
801	T104	J41	0	937	277	47	start	M06
808	T073	J32	0	1250	622	126	finish	M06
808	T105	J41	0	937	277	51	start	M06
808	T106	J41	0	937	277	63	start	M06
811	T074	J33	6	625	280	129	finish	M01
811	T083	J38	6	937	258	42	finish	M05
811	T107	J41	0	937	277	47	start	M05
815	T057	J31	1	937	622	236	finish	M03
815	T082	J38	6	937	258	46	finish	M04
815	T108	J41	0	937	277	46	start	M03
815	T109	J41	0	937	277	48	start	M04
820	T111	J42	4	6250	2188	16	submit	-
845	T099	J41	0	937	277	47	finish	M08
846	T098	J41	0	937	277	48	finish	M08
847	T096	J41	0	937	277	49	finish	M07
847	T100	J41	0	937	277	49	finish	M08
847	T110	J41	0	937	277	47	start	M07
848	T097	J41	0	937	277	50	finish	M07
848	T104	J41	0	937	277	47	finish	M06
850	T103	J41	0	937	277	52	finish	M08
856	T095	J41	0	937	277	58	finish	M03
856	T102	J41	0	937	277	58	finish	M08
858	T107	J41	0	937	277	47	finish	M05
859	T105	J41	0	937	277	51	finish	M06
861	T108	J41	0	937	277	46	finish	M03
862	T072	J32	0	1250	622	180	finish	M06
863	T109	J41	0	937	277	48	finish	M04
867	T094	J40	0	1250	777	71	finish	M07
869	T091	J40	0	1250	777	73	finish	M06
870	T090	J40	0	1250	777	74	finish	M05
871	T106	J41	0	937	277	63	finish	M06
872	T101	J41	0	937	277	74	finish	M08
872	T111	J42	4	6250	2188	16	start	M08
876	T086	J40	0	1250	777	80	finish	M05
877	T093	J40	0	1250	777	81	finish	M07
878	T087	J40	0	1250	777	82	finish	M05
878	T092	J40	0	1250	777	82	finish	M07

time	task	job	priority	CPU	memory	run time	event	machine
879	T085	J40	0	1250	777	83	finish	M03
879	T089	J40	0	1250	777	83	finish	M05
880	T084	J39	1	937	777	86	finish	M06
880	T088	J40	0	1250	777	84	finish	M05
882	T081	J38	6	937	258	113	finish	M03
888	T111	J42	4	6250	2188	16	finish	M08
894	T110	J41	0	937	277	47	finish	M07
909	T112	J43	4	6873	1910	2879	submit	-
909	T113	J43	4	6873	1910	2869	submit	-
909	T114	J43	4	6873	1910	2869	submit	-
909	T115	J43	4	6873	1910	2872	submit	-
909	T112	J43	4	6873	1910	2879	start	M05
921	T116	J44	0	3125	466	266	submit	-
921	T116	J44	0	3125	466	266	start	M03
952	T117	J45	4	6873	1910	2969	submit	-
952	T118	J45	4	6873	1910	2980	submit	-
952	T119	J45	4	6873	1910	2959	submit	-
952	T120	J45	4	6873	1910	2966	submit	-
968	T080	J38	6	937	258	199	finish	M03
981	T121	J46	0	6250	466	38	submit	-
981	T121	J46	0	6250	466	38	start	M02
982	T122	J47	1	1250	1590	374	submit	-
982	T122	J47	1	1250	1590	374	start	M03
983	T123	J48	1	2499	311	252	submit	-
983	T123	J48	1	2499	311	252	start	M06
991	T049	J26	9	4688	6360	491	finish	M05
993	T124	J49	9	6250	1590	79	submit	-
993	T124	J49	9	6250	1590	79	start	M07
997	T125	J50	4	62	398	18	submit	-
997	T126	J51	4	625	311	209	submit	-
997	T125	J50	4	62	398	18	start	M01
997	T126	J51	4	625	311	209	start	M01
1012	T127	J52	4	6873	1910	2935	submit	-
1012	T128	J52	4	6873	1910	2902	submit	-
1012	T129	J52	4	6873	1910	2940	submit	-
1012	T130	J52	4	6873	1910	2912	submit	-
1015	T125	J50	4	62	398	18	finish	M01
1016	T131	J53	1	1250	466	175	submit	-
1016	T131	J53	1	1250	466	175	start	M05
1019	T121	J46	0	6250	466	38	finish	M02
1025	T132	J54	1	3748	466	197	submit	-
1025	T132	J54	1	3748	466	197	start	M02
1028	T133	J55	6	645	197	72	submit	-
1028	T134	J56	0	1524	271	174	submit	-
1028	T135	J56	0	1524	271	175	submit	-
1028	T136	J56	0	1524	271	175	submit	-
1028	T137	J56	0	1524	271	177	submit	-
1028	T138	J56	0	1524	271	175	submit	-
1028	T139	J56	0	1524	271	177	submit	-
1028	T140	J56	0	1524	271	173	submit	-
1028	T141	J56	0	1524	271	175	submit	-
1028	T142	J56	0	1524	271	173	submit	-
1028	T143	J56	0	1524	271	176	submit	-
1028	T144	J56	0	1524	271	175	submit	-
1028	T145	J56	0	1524	271	174	submit	-
1028	T146	J56	0	1524	271	176	submit	-
1028	T147	J56	0	1524	271	173	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1028	T148	J56	0	1524	271	174	submit	-
1028	T149	J56	0	1524	271	176	submit	-
1028	T133	J55	6	645	197	72	start	M02
1028	T134	J56	0	1524	271	174	start	M02
1028	T135	J56	0	1524	271	175	start	M05
1028	T136	J56	0	1524	271	175	start	M05
1028	T137	J56	0	1524	271	177	start	M06
1028	T138	J56	0	1524	271	175	start	M06
1028	T139	J56	0	1524	271	177	start	M08
1028	T140	J56	0	1524	271	173	start	M08
1028	T141	J56	0	1524	271	175	start	M08
1028	T142	J56	0	1524	271	173	start	M08
1041	T150	J57	6	937	197	59	submit	-
1041	T151	J57	6	937	197	59	submit	-
1041	T152	J57	6	937	197	59	submit	-
1041	T150	J57	6	937	197	59	start	M04
1041	T151	J57	6	937	197	59	start	M05
1047	T003	J03	0	1688	295	996	finish	M01
1047	T152	J57	6	937	197	59	start	M01
1058	T001	J01	1	1250	777	1057	finish	M01
1058	T143	J56	0	1524	271	176	start	M01
1069	T153	J58	1	1250	777	1139	submit	-
1072	T124	J49	9	6250	1590	79	finish	M07
1072	T153	J58	1	1250	777	1139	start	M07
1072	T144	J56	0	1524	271	175	start	M07
1072	T145	J56	0	1524	271	174	start	M07
1072	T146	J56	0	1524	271	176	start	M07
1100	T133	J55	6	645	197	72	finish	M02
1100	T150	J57	6	937	197	59	finish	M04
1100	T151	J57	6	937	197	59	finish	M05
1102	T154	J59	9	4688	6360	487	submit	-
1102	T136	J56	0	1524	271	175	evict	M05
1102	T135	J56	0	1524	271	175	evict	M05
1102	T131	J53	1	1250	466	175	evict	M05
1102	T154	J59	9	4688	6360	487	start	M05
1103	T155	J60	1	625	622	783	submit	-
1103	T155	J60	1	625	622	783	start	M01
1106	T152	J57	6	937	197	59	finish	M01
1112	T156	J61	4	6873	1910	2931	submit	-
1112	T157	J61	4	6873	1910	2911	submit	-
1112	T158	J61	4	6873	1910	2994	submit	-
1112	T159	J61	4	6873	1910	2907	submit	-
1121	T160	J62	0	1688	295	1070	submit	-
1140	T008	J08	4	3381	1413	459	finish	M04
1140	T131	J53	1	1250	466	175	start	M04
1140	T135	J56	0	1524	271	175	start	M04
1140	T136	J56	0	1524	271	175	start	M04
1187	T116	J44	0	3125	466	266	finish	M03
1187	T147	J56	0	1524	271	173	start	M03
1187	T148	J56	0	1524	271	174	start	M03

Cell window 25: Events

Time 579: machine M02 failed; tasks T007, T008, T057 stopped and went back to the queue.

Time 579: task T057 restarted on machine M03.

Time 659: task T007 restarted on machine M01.

Time 681: task T008 restarted on machine M04.

Time 934: machine M02 returned.

Time 1102: task T154 (priority 9) evicted task T131 (priority 1) from machine M05.

Time 1102: task T154 (priority 9) evicted task T135 (priority 0) from machine M05.

Time 1102: task T154 (priority 9) evicted task T136 (priority 0) from machine M05.

Time 1140: task T131 restarted on machine M04.

Time 1140: task T135 restarted on machine M04.

Time 1140: task T136 restarted on machine M04.

Longest waits (submission to first start, tasks started by the window end): task T147 waited 159 s (submitted at 1028, first started at 1187); task T148 waited 159 s (submitted at 1028, first started at 1187); task T071 waited 107 s (submitted at 575, first started at 682); task T072 waited 107 s (submitted at 575, first started at 682); task T073 waited 107 s (submitted at 575, first started at 682).

Cell window 25: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T007	J07	8	1874	777	M01	659	2872
T045	J22	8	625	466	M01	391	586
T126	J51	4	625	311	M01	997	6
T143	J56	0	1524	271	M01	1058	34
T155	J60	1	625	622	M01	1103	686
T132	J54	1	3748	466	M02	1025	22
T134	J56	0	1524	271	M02	1028	2
T012	J09	1	1250	777	M03	223	117
T122	J47	1	1250	1590	M03	982	156
T147	J56	0	1524	271	M03	1187	160
T148	J56	0	1524	271	M03	1187	161
T016	J13	0	1688	295	M04	271	105
T131	J53	1	1250	466	M04	1140	115
T135	J56	0	1524	271	M04	1140	115
T136	J56	0	1524	271	M04	1140	115
T112	J43	4	6873	1910	M05	909	2588
T154	J59	9	4688	6360	M05	1102	389
T123	J48	1	2499	311	M06	983	35
T137	J56	0	1524	271	M06	1028	5
T138	J56	0	1524	271	M06	1028	3
T144	J56	0	1524	271	M07	1072	47
T145	J56	0	1524	271	M07	1072	46
T146	J56	0	1524	271	M07	1072	48
T153	J58	1	1250	777	M07	1072	1011
T139	J56	0	1524	271	M08	1028	5
T140	J56	0	1524	271	M08	1028	1
T141	J56	0	1524	271	M08	1028	3
T142	J56	0	1524	271	M08	1028	1

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T113	J43	4	6873	1910	909	2869
T114	J43	4	6873	1910	909	2869
T115	J43	4	6873	1910	909	2872

task	job	priority	CPU	memory	submitted	run time
T117	J45	4	6873	1910	952	2969
T118	J45	4	6873	1910	952	2980
T119	J45	4	6873	1910	952	2959
T120	J45	4	6873	1910	952	2966
T127	J52	4	6873	1910	1012	2935
T128	J52	4	6873	1910	1012	2902
T129	J52	4	6873	1910	1012	2940
T130	J52	4	6873	1910	1012	2912
T156	J61	4	6873	1910	1112	2931
T157	J61	4	6873	1910	1112	2911
T158	J61	4	6873	1910	1112	2994
T159	J61	4	6873	1910	1112	2907
T149	J56	0	1524	271	1028	176
T160	J62	0	1688	295	1121	1070

All machines are up at time 1200.